Persuasion with Partisanship: The Informational Content of Policymaking with Application to U.S. Governors

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Abstract

I study a model where a political executive's policy agenda generates information about her ability. Since partisan policies are harder to pass, an agenda's partisanship affects what success or failure communicates about ability. The model predicts a U-shape between office-motivated executives' popularity and their agendas' partisanship. I then analyze the partisanship of U.S. governors' policy proposals from 1990-2020 using a novel text dataset. Despite a rise in partisanship over the sample period, I argue that electoral incentives can explain substantive variation in partisanship, testing the model and showing a U-shape between partisanship and approval as well as other comparative statics.

JEL Codes: D72, D73, D78, D83

Keywords: partisanship, persuasion, policy learning, polarization, reputation, accountability, governors, state politics, executives

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1 Introduction

Scholars have documented a rise in partisan political behavior in the U.S. Congress since the 1990s that is apparent in political speeches, voting behavior, and policy proposals.[1] However, over this same time period, the executive branch has increasingly dominated policymaking, particularly at the state level.[2] On the one hand, state governors have increasingly spearheaded highly partisan agendas across the states, ranging from Republican Ron DeSantis's expansive immigration enforcement initiatives (DeSantis, 2026) to Democrat Tim Walz's progressive efforts to codify abortion access and accelerate transitions to clean energy (Masters, 2024). On the other, many governors pursue agendas that cross partisan lines. Republican Mitt Romney's expansion of health insurance in Massachusetts ("Romneycare") has been widely regarded as a precursor to the Affordable Care Act (Belluck, 2006). More recently, Republican Kelly Armstrong expanded childcare subsidies in North Dakota (Steurer, 2025), while Democrat Jared Polis controversially adopted a pro-school-choice tax credit offered by the Trump administration (Mahnken, 2026).

Despite the national increase in partisanship, what drives variations in the partisanship of governors' policy agendas? One answer may be that executives strategically choose policies to convey information about their *political skills*. Voters may positively view a governor who convinces a divided legislature to pass a radical, partisan policy, or negatively view a governor who fails to pass a bipartisan policy most legislators agree is a good idea. A governor teetering on the edge of winning reelection may hence benefit from prioritizing different policies than one who is very likely to win (or lose) reelection.[3] Policy agendas can thus be a tool for politicians to *persuade* voters of their political skills, with the informativeness of those agendas depending on their partisanship.

This paper provides evidence for this hypothesis by coupling a theoretical model of persuasion with novel empirical evidence from U.S. governors' policy agendas. I develop a model where an incumbent's policy agenda generates an *information structure* mapping her political ability to a likelihood of policy success. Partisan policies that appeal only to the incumbent's party require more skill to pass, meaning their failure can be attributable to legislative opposition. Bipartisan policies that appeal to the opposition require less skill to pass, meaning their failure signals low ability. The model predicts a U-shaped relationship between an incumbent's popularity and the partisanship of her agenda. Unpopular incumbents pursue partisan agendas to salvage reelection. Popular incumbents propose partisan policies whose failure can plausibly be blamed on legislative antagonism. Incumbents in the middle pursue safe bipartisan policies to secure reelection. The U-shape weakens as the incumbent's alignment with the legislature increases.

[1] See McCarty (2019) for an overview of the forces driving trends in partisanship.

[2] Andeweg, Elgie, Helms, Kaarbo, and Müller-Rommel (2020) discuss the reemergence of executive power since the mid-20th century, as well as the expanding role of the U.S. Presidency and its administrative apparatus in shaping legislative discourse. Goldgeier and Saunders (2018) note a rise in the executive branch's control over foreign policy, which has become increasingly left unchecked by Congress. Reynolds (2024) and E. Peterson (2024) provide stylized overviews of the eroding power of Congress over the executive branch. The burgeoning power of U.S. state governors, especially after the 1990s, is discussed by Heidbreder (2012).

[3] Chapter 4 of Rosenthal (2012), for instance, discusses how governors' policy agendas are strategically chosen by the composition of the legislature and what will pass.

I then study the partisanship of U.S. gubernatorial policy agendas and test the predictions of the model on a text dataset of governors' policy proposals. I compile a panel of governors' annual "State of the State" addresses to the legislature. I utilize large-language modeling to isolate text related to policy proposals. I measure the partisanship of these proposals via divergence in phrases used by Democrats and Republicans. I show that while aggregate gubernatorial partisanship has steadily increased since the early 2000s, it is lower than that estimated from the Congressional Record and that there are substantial deviations from these trends at the governor level. [4]

With this heterogeneity in mind, I turn to the panel data to test the model's predictions. I compare the partisanship of reelectable governors' proposals to those of term-limited governors ineligible for election. This controls for forces that influence partisanship of both term-limited and reelectable governors — such as preferences for partisan policies — and thereby isolates variations in partisanship emerging from electoral concerns. I show a nonmonotonic relationship between gubernatorial approval and partisanship of policy proposals for reelectable governors, in line with the model's prediction. This relationship weakens as legislative alignment increases.

Theoretical Model The baseline model considers an incumbent executive of unknown ability who chooses a policy agenda of varying partisanship. I interpret ability as an executive's managerial skill, capacity to follow through on promises, or valence, and refer to expected ability as *reputation*. An agenda's probability of passing is decreasing in its partisanship but increasing in the incumbent's ability. This captures the notion that partisan policies appealing to a single party are harder to garner support for than bipartisan compromises amenable to most of the legislature, which even low-skilled politicians may pass. A representative voter observes the incumbent's choice of agenda, whether it passes or fails, updates beliefs over the incumbent's ability, and reelects her if and only if her reputation is sufficiently high. Incumbents are office-motivated, meaning the model captures the behavior of reelectable relative to term-limited incumbents.

The key idea of the model is that an incumbent's choice of policy is, in fact, also a choice of information structure, as in models of persuasion. Given a prior over the incumbent's ability, the success or failure of an agenda generates a distribution of posteriors over ability whose skewness is determined by the agenda's partisanship. The incumbent chooses this distribution — the partisanship of her policy agenda — to maximize chances of reelection. The informativeness of success or failure is disciplined by the incumbent's alignment with the legislature. Intuitively, an incumbent aligned with her legislature has power over the government and receives credit or blame for policy successes or failures. An incumbent unaligned with her legislature can often blame policy failures on an uncooperative legislature.

The baseline model highlights a fundamental nonmonotonicity in how partisan the incumbent's policy agenda is as her reputation increases. This is the core theoretical prediction of the model. Partisan policies are likely to fail, but reveal a high ability incumbent if they succeed.

[4] Specifically, the level of gubernatorial partisanship is below that of the Congressional level estimated by Gentzkow, Shapiro, and Taddy (2019).

Bipartisan policies are likely to succeed, but reveal a low ability incumbent if they fail. Hence, low reputation incumbents likely to lose reelection pursue partisan policies to save their chances of retaining office. Middling reputation incumbents pursue low-risk bipartisan policies to secure reelection. High reputation incumbents pursue partisan policies whose failure can be blamed on legislative antagonism, meaning the incumbent's reputation falls by less. Therefore, the model delivers a U-shaped relationship between an incumbent's reputation and her agenda's partisanship.

This U-shape flattens out if legislative alignment increases. Decreasing electoral competition also makes the U-shape less salient — for example, in constituencies where an incumbent's party always wins reelection. I then show that the results of the model extend to settings where a high ability politician may pursue legislation that voters are ideologically misaligned with. In equilibrium, voters "check" an incumbent's powers by choosing a legislature that is less aligned with her, thereby dampening the negative effects of ideology. I use this parametrization of voter utility to show that voter welfare is generally nonmonotonic in reputation.

After presenting the baseline model, I extend it by allowing endogenous differentiation of policy passage by ability. This generalization renders the model similar to a Bayesian Persuasion problem where a sender cannot totally obfuscate information; in this case, the incumbent cannot pursue a policy agenda whose success or failure is totally uninformative of her ability. [5] The mathematical flexibility of the general approach allows me to add a preference shock and write the model's predictions in terms of ex-ante *winning probability*.

I show that the partisanship of the incumbent's policy agenda indeed exhibits a U-shape as a function of ex-ante winning probability, which flattens when legislative alignment increases or competition weakens. Since proxies for politicians' winning probability — approval ratings — are readily available, the generalization delivers testable predictions of the theory: the partisanship of an executive's policy agenda should follow a U-shape as a function of her approval rating and flatten under high legislative alignment or low electoral competition.

The general model shows that the U-shape relationship between reputation and partisanship may in fact be asymmetric. Popular incumbents may pursue *more partisan* agendas than unpopular incumbents, since low reputation incumbents prefer policies that differentiate high and low-ability incumbents, while high reputation incumbents prefer policies whose failure is as uninformative as possible. It also allows me to show that the model's key predictions are robust to multiple periods of agenda passage or allowing incumbents to choose policies from the opposing party.

Documenting Gubernatorial Partisanship I apply the insights of the model to explain variation in the state-level partisanship of U.S. governors' policy agendas. I begin by assembling a corpus of U.S. governors' "State of the State" speeches. These addresses to the state legislature are given in the first quarter of each year and lay out the governor's policy proposals for the coming

[5] In particular, the model can be seen as a Bayesian Persuasion problem where a receiver (voter) is uncertain whether he observes a signal because it was endogenously chosen by a sender (incumbent) or whether it was generated by an outside source (i.e. whether a policy outcome is a consequence of the incumbent's inherent skill or the legislature's involvement).

year. I collect over 800 speeches and merge them with existing data to assemble an annual panel of over $1,300$ speeches from 1990-2020. Because these speeches also contain reflections on past policies and rhetoric, I fine-tune a large-language-model called "BERT" (Devlin, Chang, Lee, & Toutanova, 2018) to isolate speech related to policy proposals. I construct an aggregate measure of partisanship of policy proposals for each year and a panel measure for each governor-year by measuring divergence of phrase-use between Republican and Democratic governors using Gentzkow, Shapiro, and Taddy (2019).

My aggregate series shows a steady rise in partisanship of policy proposals beginning in the early 2000s that finishes with a spike in 2017.[6] Gubernatorial partisanship is consistently lower than Congressional partisanship, and the timing of changes is different. Additionally, I show that the phrases identified as "most partisan correspond to concrete policy proposals associated with both Republicans and Democrats. Republicans tend to discuss taxation, tax cuts, charter schools, and crime, while Democrats tend to discuss healthcare, the minimum wage, and topics related to energy. Nevertheless, I show that slower-moving aggregate trends disguise sizable variation in partisanship at the governor level, such as spikes in the late 1990s surpassing the levels of the late 2010s and dips in the mid-2010s lower than those in the 1990s.

Testing Model and Implications I then use the gubernatorial panel to explain variation in the partisanship of policy proposals through the lens of my model. First, I utilize data from Kousser and Phillips (2012) to show that many of the central assumptions of the theory are borne out in the data. Proposals from more partisan policy agendas are consistently less likely to pass: in particular, a one standard deviation increase in partisanship means a proposal is up to nine percentage points less likely to pass. Additionally, higher approval governors are more likely to pass their agendas, and this is primarily driven by governors aligned with their legislatures; both predictions are consistent with the theory.

Next, I test the central behavioral predictions of the model by looking at the difference in the partisanship of reelectable and term-limited governors' agendas. To capture the effects of winning probability and legislative alignment, I interact deciles of gubernatorial approval ratings with an indicator for whether the governor's party controls the state legislature. I study how the governor-year measure of partisanship moves with changes in these variables. To highlight how electoral incentives drive variation in partisanship, I compare the partisanship of reelectable and term-limited governors, thereby controlling for governor-level policy preferences or time trends that drive partisanship but impact both groups.

I show that when comparing reelectable to term-limited governors, partisanship of proposals exhibits a U-shape as a function of approval decile for governors unaligned with their legislatures. The speech of high or low approval governors is over one standard deviation higher relative to middle approval governors. The U-shape is weaker for governors aligned with their legislatures and states where Democratic governors consistently win elections. The findings are robust to

[6]These trends are identical when estimated on the full corpus of governor speech.

alternative definitions of legislative alignment, different permutations of election and non-election years, governor-level fixed effects, and controlling for governors' presidential aspirations.

These insights suggest that partisanship should be viewed not as a slow-moving aggregate but as a dynamic force shaped by electoral and strategic concerns, popularity, and legislative alignment. They offer a reputational explanation for why executives compromise with the opposition, assign credit or blame for policy outcomes, or pursue radical partisan reforms.

Literature At a high level, this paper develops a model of persuasion to show how career concerns can drive variation in partisanship. It then formally tests the predictions of that model on U.S. governors' policy agendas.

This paper contributes to a theoretical literature on political career concerns by developing a model that exhibits a nonmonotonic relationship between policy extremity and reputation. Papers establishing a negative monotonicity known as "gambling for resurrection" (Dur, 2001; Fu & Li, 2014; Izzo, 2024; Majumdar & Mukand, 2004) cannot explain the nonmonotonicities evident in the gubernatorial data on partisan policy proposals. Similarly, the mechanism of the model is not driven by pandering or voter inattention to policy choices, unlike Canes-Wrone, Herron, and Shotts (2001) or Maskin and Tirole (2004).[7] The paper's welfare insight on the ambiguous effects of electoral competition is also related to Dewan and Hortala-Vallve (2019), is studied in the settings of electoral accountability by Ashworth, De Mesquita, and Friedenberg (2017) and Bils and Izzo (2022), and is applied to executives' willingness to pursue unilateral actions in Judd (2017). These papers show how equilibrium outcomes are sensitive to parameters of competition, office rents, and voter beliefs. My model predicts a singular U-shaped relationship between partisanship and popularity that shifts or flattens with such changes.

The present paper provides insights into solving a constrained Bayesian Persuasion problem, following Kamenica and Gentzkow (2011), where a receiver is uncertain about the origin of a sender's signal. The key constraint on the sender (incumbent) in my environment is that she is unable to implement totally uninformative information structures (policy agendas). To this end, the model is related to a literature on Bayesian Persuasion with mediation, multiple senders, and cheap talk (Alonso & Câmara, 2016; Arieli, Babichenko, & Sandomirskiy, 2022; Ichihashi, 2019; Lipnowski, Ravid, & Shishkin, 2022). The present paper's model – where the difficulty of passing more partisan policies produces skewed information about an executive's ability — is connected to models of information design in test-taking, such as Hancart (2024). More broadly, the present paper also *tests* a model of persuasion in a high-stakes, political economy setting.

I contribute to the empirical literature on political accountability by comprehensively linking ideological partisanship to legislative alignment, popularity, and reelection eligibility. Prior papers study such topics as the effects of term limits and career-concerns on partisanship (Besley & Case, 1995; Iaryczower, Lopez-Moctezuma, & Meirowitz, 2024); the effect of reelection on wel-

[7] Canes-Wrone et al. (2001) assume executives want to both win reelection and implement the optimal policy. The latter preference drives non-pandering in edge cases of their model, which is absent if politicians only want to win reelection.

fare reform (Bernecker, Boyer, & Gathmann, 2021) or COVID-19 stringencies (Pulejo & Querubín, 2021); or the economic returns to holding a Congressional seat (Diermeier, Keane, & Merlo, 2005).

I add to research on partisanship of political speech by providing a novel, unidimensional series on partisanship of political executives' speech and partisanship at the state level. Series for partisanship of speech have been studied in the U.S. Congressional Record (Gentzkow, Shapiro, & Taddy, 2019; Jensen et al., 2012) and other countres like the UK (A. Peterson & Spirling, 2018), but the few studies at the state level have been limited to counts of pre-selected words or nationalization trends (Butler & Sutherland, 2023; Hopkins, Schickler, & Azizi, 2022). I also contribute to a literature analyzing State of the State speeches, which I discuss in the data section.

My utilization of advances in large-language-modeling techniques adds to a growing literature in Economics utilizing LLMs, including applications to labor contracts and the FOMC (Arold, Ash, MacLeod, & Naidu, 2024; Hansen, McMahon, & Prat, 2018), which are reviewed by Gentzkow, Kelly, and Taddy (2019) and Ash and Hansen (2023). I utilize the same methods as Card et al. (2022), who fine-tune two models to isolate speeches related to immigration in the Congressional Record and then categorize those speeches by sentiment.

Finally, I complement a literature on the mixed (and heterogeneous) behavior of partisanship in state and local politics utilizing roll-call data, close elections, and other non-text measures. Some studies argue that governor and mayoral party identity has little impact on partisanship of most political outcomes (Ferreira & Gyourko, 2009; Leigh, 2008), while others argue for modest or larger effects of party control on partisanship after the 1990s, measured using liberalism scores; law passage by policy domain; taxation, debt, housing stock, and public expenditures; or policy diffusion across states (Carlino, Drautzburg, Inman, & Zarra, 2023; Caughey, Xu, & Warshaw, 2017; de Benedictis-Kessner, Jones, & Warshaw, 2024; de Benedictis-Kessner & Warshaw, 2020; DellaVigna & Kim, 2022; Grumbach, 2018). Heterogeneous increases in partisanship of legislative voting behavior at the state level have been documented by Shor and McCarty (2011) and updated by DellaVigna and Kim (2022). By viewing partisan variation as products of incumbents' electoral incentives, my paper provides structure to these more heterogeneous outcomes from the standpoint of executives. To this end, the present paper differs from attempts to measure partisanship of inherent gubernatorial ideology, such as the campaign finance DIME measure constructed by Bonica (2014) and evaluated for governors in Warner (2023).

Plan of the Paper I introduce the theoretical model in the next section. I then solve the baseline model, show robustness to ideological misalignment with voters, solve the general model with a valence shock, and detail data predictions. I next introduce the data, explain the relevance of the model to the gubernatorial setting, and detail estimation of partisanship. Next, I study how aggregate trends in partisanship mask substantial heterogeneity in the panel data. With this heterogeneity in mind, I show how the predictions of the model explain variation in gubernatorial partisanship and then conclude.

2 Theoretical Model

I begin by introducing a benchmark model highlighting the key intuitions of the theory. After showing the central result, I first show that the model's key insights are robust to polarized envirnoments where a voter values a politician's ideology and use this to study voter welfare. I then move to an extended model that maintains the key intuitions of the benchmark while permitting mathematical flexibility to solve the model with a valence shock. This allows me to write the model's comparative statics in terms of the incumbent's win probability and thereby utilize approval ratings as an independent variable when testing the model's comparative statics in the empirical section. I finish by addressing two extensions to the model.

2.1 Preliminaries

Setup There are two time periods. At $t = 1$, an incumbent politician R (she) holds office. R chooses a policy agenda $\pi \in [0, 1]$, which may pass or fail. Higher π represents a more partisan agenda that favors R's party. The probability of passage is increasing in R's ability $a_R \in \{0, 1\}$, which is unknown to all agents.[8] Higher ability politicians are more effectively able to pass and/or implement legislation.

A representative voter V (he) observes R's choice of agenda, whether it passes or fails, and updates his beliefs about R's ability. Then, at $t = 2$, V chooses to retain R or replace her with a challenger L of unknown ability $a_L \in \{0, 1\}$. q_i^t is the belief politician i is high ability ($a_i = 1$) at time t, which we refer to as i's "reputation." V's utility is the ability a_i of politician i in office. Politicians are office-motivated, receiving a payoff of 1 upon reelection and 0 otherwise.

Policies The success of R's policy agenda is a function of its partisanship $\pi \in [0, 1]$; R's ability a_R; and the legislature's alignment with R, $\lambda \in [0, 1]$, which represents the relative strength of R's party in the legislature. The probability of passage is:

$$p(\lambda, a_R, \pi) = \lambda a_R + (1 - \lambda)(1 - \pi).$$

Legislative alignment λ dictates the relative control of the incumbent over the legislature and, relatedly, how much blame can be placed on the incumbent for a success or failure. When $\lambda = 1$, the legislature is totally aligned with the incumbent. Success or failure are entirely driven by R's ability. When $\lambda < 1$, the legislature involves itself in the passage of policies so that passage is a weighted combination of R's ability and her agenda's partisanship. Pursuing the least partisan agenda $\pi = 0$ always leads to success when $a_R = 1$ and may still succeed when $a_R = 0$. Pursuing the most partisan agenda $\pi = 1$ succeeds only if $a_R = 1$, and even then may still fail. For $\lambda = 0$, only partisanship affects whether an agenda passes.

The choice of $\pi \in [0, 1]$ then amounts to the choice of an *information structure*. That is, the

[8] The model's equilibrium is robust to an incumbent privately knowing her type.

agenda maps a state (the incumbent's underlying ability) to a distribution of signals (success or failure of that agenda). The voter infers the incumbent's type from these signals and uses this to update his beliefs over how competent the incumbent is.

Comments Because incumbents are office-motivated, the model should be thought of as isolating variation in partisanship due to reelection incentives. Adding flow utility over partisan policies (e.g. due to an incumbent's preferences or external constraints) would add a unidirectional preference for more partisan policies. Because such forces would also operate if an incumbent were term-limited, the predictions of the model should emerge when comparing reelection-eligible to term-limited incumbents. This comparison controls not only for executives' preferences but also contemporaneous shocks that affect the partisanship of both reelectable and term-limited executives equally, and will serve as a cornerstone of the empirical tests of the model.

The model interprets ability as a politician's managerial skill, which may include efficacy in crafting and implementing laws, managing bills and the budget, and maintaining a smooth-running bureaucracy. Governors themselves often use the analogy of a "CEO" to describe the duties and skills ascribed to their job (Behn, 1991). Ability can also represent a capacity to commit to following through on a policy agenda; or the incumbent's valence, which is higher if she is perceived to be better at passing legislation. The data section provides examples of these skills from the National Governors Association, noting that the success or failure of a policy agenda is one of the most crucial inputs into voters' perceptions of governors.

The baseline model assumes that the representative voter V prefers a high ability incumbent to a challenger of unknown ability to a low ability incumbent. This assumption may capture two sorts of phenomena. First, V may be a member of R's party and, all else equal, prefers that the executive of her party is high ability. The second interpretation is that V is a median voter who has horizontal preferences (over a policy space) and vertical preferences (over a politician's ability). All else equal, he values R's managerial ability to address nonpartisan tasks of governing. That the latter channel is valuable regardless of a median voter's ideological preferences is empirically consistent with the fact that many "Republican" states tend to have governors who are Democrats and vice-versa. Section further explores the interaction of ability and ideology and shows that the model is robust to these considerations.

All else equal, bipartisan policies are more likely to pass than partisan policies. In many polarized electoral environments, passing even bipartisan legislation may be seen as a success. What matters for the analysis is that even if passing bipartisan policies is a strong signal of political skill, passing even *more* partisan policies is an even stronger signal. Bipartisan policies should be thought of as *ideologically* bipartisan (or even nonpartisan) policies, as opposed to bills requiring bipartisan *support* to pass. [9] The behavior of the legislature in the model is relatively passive:

[9] Curry and Lee (2020) provide a comprehensive picture of the landscape of (bi)partisan bill passage in the U.S. Congress. One Congressional staffer notes that "[a] bill still has a lot greater chance to make it into law if it's bipartisan" (p. 44), while another states that "[g]etting anything done is hard, but it's even harder on a partisan basis" (p. 45). Another staffer points out that while some bipartisan legislation may be passed with easy support from both parties,

legislators are more likely to pass policies with which they are more aligned. This assumption is motivated by studies like Lebo and O'Geen (2011), which show that parties are incentivized to see their executive's policy agenda through because it makes them more likely to gain legislative seats. This behavior is also consistent with settings where voters choose legislators who are more likely to express their preferences (Ansolabehere & Kuriwaki, 2022; Fouirnaies & Hall, 2022; Jones & Walsh, 2018; Lee, Moretti, & Butler, 2004).

The model's equilibrium is robust to R privately knowing her type. The intuition is that actions are not hidden: the voter observes R's *choice* of π and whether it passed or failed. If high and low ability incumbents chose different policies, voters would instantly infer their type, meaning the two must pursue the same policy in equilibrium. This differentiates the model from career concerns models like Holmstrom (1999), where choices are unobservable.

In the model, independently of the incumbent's policy choice, some information is bound to be revealed about her type with positive probability. The motivation is that incumbents oversee programs run by the executive branch, basic functioning of government, and passage of "business as usual" legislation. Relatedly, the assumption of a binary type space (high/low ability) is largely without loss, as R is reelected if and only if her *expected* ability is sufficiently high. Adding "intermediate" types would not change the fundamental results as long as — all else equal — higher ability types are better at passing legislation.

The model defines partisanship as *relative* to a legislature's composition. Suppose λ increases to λ'. A bipartisan policy for the λ' legislature will likely be more partisan than a bipartisan policy for the λ legislature. π, to this end, identifies partisanship relative to a base level for the legislature.

Voters in the model can learn about an incumbent but not the challenger, meaning the expected ability of a challenger is effectively an outside option for voters. I take the expected ability of the challenger to generally describe how electorally competitive the incumbent's jurisdiction is. Voters learn about the incumbent's ability precisely because she is in office.

2.2 Equilibrium

The voter elects whichever politician has higher expected ability. R wins reelection if and only if $q_R^2 \geq q_L^2$. Hence, the equilibrium utility of R as a function of q_R^2 is $u_R(q_R^2) \equiv \mathbf{1}[q_R^2 \geq q_L^2]$.

Information Given a prior q_R^1, the partisanship of each policy agenda generates posteriors q_R^2 about R's ability, given by $F(q_R^2|q_R^1, \pi)$. In the setting with success/failure, we end up with either a posterior higher than q_R^1 (denoted $\overline{q}_R^2$) or lower than q_R^1 (denoted $\underline{q}_R^2$). The incumbent's general

more ambitious legislation is only able to pass with negotiation between party leaders and veto-holders. This suggests a level of persuasion skill or political acumen to secure passage of more partisan initiatives. Despite increasing party polarization, the authors show that the sorts of bipartisan support and tactics required to secure legislation is empirically no different now than in past decades.

problem is then given by:

$$V_R(q_R^1) = \max_{\pi \in [0,1]} \int_{q_R^2} u_R(q_R^2) dF(q_R^2 | q_R^1, \pi). \tag{1}$$

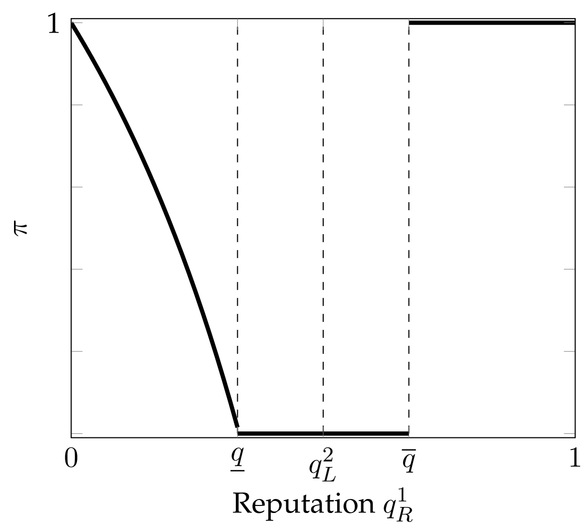
Let $\pi^*(q_R^1)$ represent the solution of the program above. The following proposition describes π^*.

Proposition 1. *There exist thresholds* $0 < \underline{q} < q_L^2 < \overline{q} < 1$ *such that:*

- *for* $q_R^1 \in [0, \underline{q})$, $\pi^*(q_R^1)$ *is strictly decreasing, with* $\pi^*(0) = 1$ *and* $\pi^*(\underline{q}) = 0$;
- *for* $q_R^1 \in [\underline{q}, \overline{q})$, $\pi^*(q_R^1) = 0$;
- *for* $q_R^1 \in [\overline{q}, 1]$, $1 \in \pi^*(q_R^1)$ *with equality at* $\overline{q}$.

The solution is graphed below:

Figure 1: Partisanship of Optimal Policy π^*



All proofs are contained in Appendix A. The intuition for this result is best viewed through comparing the payoffs from $\pi = 0$ and $\pi = 1$. When $\pi = 0$, observing a success generates the posterior over ability $\overline{q}_R^2 = \frac{q_R^1}{q_R^1 + (1-\lambda)(1-q_R^1)}$. Observing a failure generates the posterior $\underline{q}_R^2 = 0$, since failure to pass bipartisan policies is damning evidence that $a_R = 0$.

When $\pi = 1$, observing a success is confirmatory evidence that $a_R = 1$, so $\overline{q}_R^2 = 1$. Because high ability incumbents may fail to pass partisan policies, the posterior upon observing failure is $\underline{q}_R^2 = \frac{(1-\lambda)q_R^1}{1-\lambda q_R^1}$. Hence, the expected winning probabilities from pursuing each policy are:

$$\begin{aligned}
\pi = 1: \qquad & (q_R^1 + (1-\lambda)(1-q_R^1))\mathbf{1}\left[\frac{q_R^1}{q_R^1 + (1-\lambda)(1-q_R^1)} \geq q_L^2\right], \\
\pi = 0: \qquad & \lambda q_R^1 + (1-\lambda q_R^1)\mathbf{1}\left[\frac{(1-\lambda)q_R^1}{1-\lambda q_R^1} \geq q_L^2\right].
\end{aligned}$$

Consider the following cases:

- Suppose q_R^1 is so low that $\frac{q_R^1}{q_R^1+(1-\lambda)(1-q_R^1)} < q_L^2$. Then R never wins if she pursues $\pi = 0$. By contrast, if she pursues $\pi = 1$, she wins with positive probability, since successes are confirmatory news of high ability. Hence, $\pi = 1$ dominates $\pi = 0$.

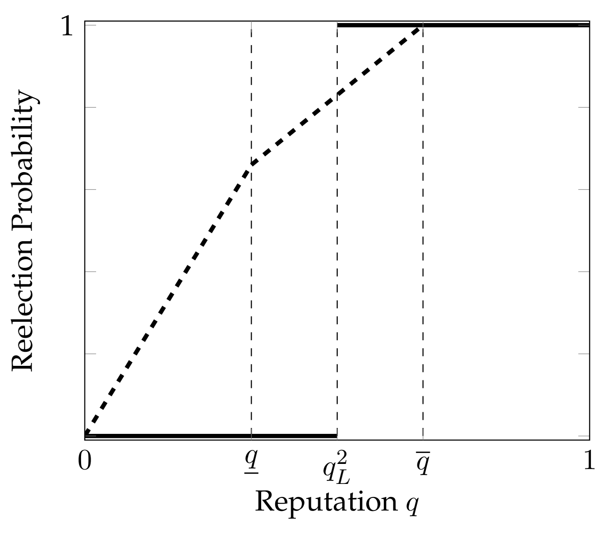
- Suppose q_R^1 is larger so that $\frac{q_R^1}{q_R^1+(1-\lambda)(1-q_R^1)} \geq q_L^2$ but such that $\frac{(1-\lambda)q_R^1}{1-\lambda q_R^2} < q_L^2$. The expected win probability from pursuing $\pi = 0$ is $q_R^1 + (1-\lambda)(1-q_R^1)$, whereas the expected win probability from pursuing $\pi = 1$ is only λq_R^1. Hence $\pi = 0$ dominates $\pi = 1$.

- Finally, suppose q_R^1 is high, so that $\frac{(1-\lambda)q_R^1}{1-\lambda q_R^1} \geq q_L^2$. Pursuing $\pi = 1$ then leads to a win with probability 1, which dominates $\pi = 0$.

Finally, note that for $q_R^1 \geq \overline{q}$, π^* may contain other values besides $\pi = 1$ because win probability is flat above q_L^2. This is an artificat of the simple model, and can be broken (in favor of $\pi = 1$) by adding a valence shock so that $u_R(q_R^1)$ is weakly concave above q_L^2 — as in the more general model presented in Theorem 1.[10]

Figure 2 shows R's equilibrium reelection odds in the first period (dashed) and the second period (solid) as a function of her reputation. Reelection in the second period follows a threshold rule. R's strategic choice of a policy agenda generally provides an incumbency advantage. Incumbents who pursue more partisan policies with $q_R^1 \leq \underline{q}$ accrue large gains, as do those who pursue bipartisan policies on $(\underline{q}, q_L^2]$. Incumbents with $q_R^1 \in (q_L^2, \overline{q})$ would prefer to pursue uninformative policies but cannot given the information structures at their disposal, and are worse off.

Figure 2: R's Ex-Ante Probability of Reelection in Second Period (Solid) vs. First Period (Dashed)



The proof of the baseline result yields the following comparative statics.

[10]Relatedly, it might seem counterintuitive that a bad politician has a worse chance of passing a bipartisan policy if she has stronger control of the legislature; this is also an artifact of the simple model, and does not necessarily hold when considering the more general information structure below.

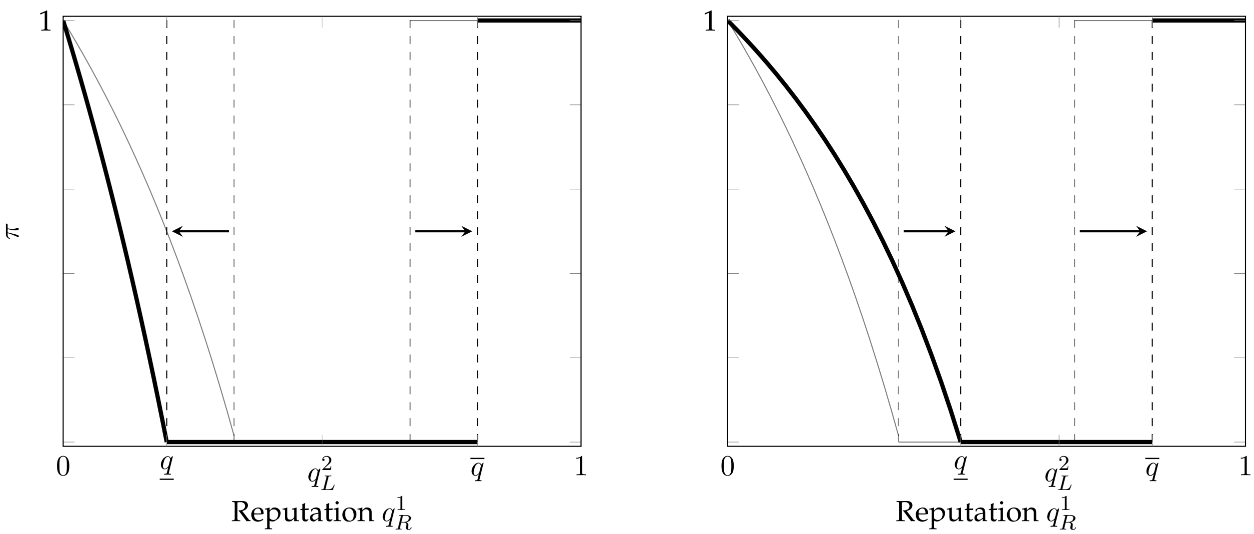
Corollary 1. *$\overline{q}$ and $\underline{q}$ vary as follows with λ:*

- *$\underline{q}$ is decreasing in λ; as $\lambda \to 0$, $\underline{q} \to 0$; as $\lambda \to 1$, $\underline{q} \to q_L^2$.*
- *$\overline{q}$ is increasing in λ; as $\lambda \to 1$, $\underline{q} \to 1$; as $\lambda \to 0$, $\overline{q} \to q_L^2$.*

$\overline{q}$ and $\underline{q}$ are both increasing in q_L^2. As $q_L^2 \to 0$, both terms $\to 0$. As $q_L^2 \to 1$, both terms $\to 1$.

Recall that $1 - \lambda$ also measures R's capacity to blame outcomes on the legislature. As λ increases, this obfuscatory power decreases, meaning that the failure of partisan policies is a more negative signal of ability. Bipartisan policies $\pi = 0$ are the "safest" in the sense that they have a high probability of success; and so, incumbents lean more on bipartisan policies for their relative safety. As q_L^2 increases, the electoral environment becomes more competitive. This causes a "right shift" in the incentives of the model, as R requires more partisan policies at the low end of q_R^1 to signal sufficiently high ability. The opposite holds in *less* competitive environments.

Figure 3: Variation of Optimal Policy π^* with λ, q_L^2



(a) Increase in Leg. Alignment λ

(b) Increase in Competitition q_L^2

2.3 Ideology and Ability

The analysis so far has taken as a given that V elects R over L as long as her reputation is sufficiently high. However, in modern environments where ideological identity seems central to voter behavior, a politician's "vertical" managerial ability may matter less than her "horizontal" ideological orientation. While the results above continue to hold as long as a politician's vertical and horizontal attributes are independent, the two may interact. For example, a skilled, partisan politician may also be skilled at passing partisan legislation that hurts voters.

This subsection extends the model to include ideology. It shows that, in equilibrium, the model's incorporation of legislative alignment preserves the central U-shape mechanism of the model. Intuitively, if a voter is highly misaligned with an incumbent but finds her ability even marginally valuable, she can stack the legislature against this politician during the second-period election, stymying her ability to pass damaging, partisan legislation once reelected. [11] Finally, the extension's parametrization of voter utility allows us to assess *first period* voter welfare, which is provided in Figure 4.

Model with Ideology Consider two dimensions of utility which are functions of ability. The first is a managerial component; for politician R, the voter's utility from this component is a_R. The second dimension is an ideological component, weighted by $\delta > 0$. The median voter has a blisspoint $b \in [0,1]$. Her ideological utility from a policy π is $-\delta|b - \pi|$.

At the beginning of $t = 2$, the voter first chooses whether to reelect R. Additionally, she chooses R's alignment with the legislature $\lambda \in [0,1]$. We assume that, if reelected, R pursues a policy $\pi_R \in [0,1]$ in the second period, with $\pi_R \geq b$. Symmetric analysis applies to $\pi_R < b$. As π_R grows further from b, the incumbent is more extreme (and misaligned) with the median voter.

The policy π_R succeeds with probability $p(\lambda, a_R, \pi)$, as above. If the incumbent's policy fails, the legislature implements a policy $\lambda\pi_R$. A fully aligned legislature ($\lambda = 1$) implements $\pi = \pi_R$ and an unaligned one ($\lambda = 0$) implements $\pi = 0$. Conditional on reelecting R, the voter's choice of legislative alignment λ^* is given by the solution to

$$U_R^2(a_R, \pi_R) = \max_{\lambda \in [0,1]} a_R - \delta\big(p(\lambda, a_R, \pi_R)|\pi_R - b| + (1 - p(\lambda, a_R, \pi_R))|\lambda\pi_R - b|\big).$$

A similar equation can be used to describe U_L^2, the utility from electing the challenger L, which is effectively an outside option.

In the original model, R's reelection odds are higher if voters value $a_R = 1$ more than $a_R = 0$. The following proposition establishes when this holds in the more general case.

Proposition 2. *Suppose the voter can choose legislative alignment λ, and his second period utility is given by $U_R^2(a_R, \pi_R)$. Then, U_R^2 is increasing in a_R for $\pi_R > b$ if and only if*

$$\delta \leq \frac{\pi_R}{\pi_R b - b^2} \equiv \overline{\delta}.$$

Moreover, $\overline{\delta} \geq \frac{4}{\pi_R}$.

If b is close to R's second period policy π_R, there is minimal ideological misalignment, and V always prefers a higher ability governor. As π_R increases relative to b, $\overline{\delta}$ falls, and the voter is made worse off by a more extreme politician. $\overline{\delta}$ achives a minimum at $\pi_R = 1$.

What is less intuitive is that for the voter to prefer a *low ability* to high ability executive, he must value ideology more than four times as much as managerial ability — i.e. $\overline{\delta} \geq \frac{4}{\pi_R}$. This is because a voter can use legislative alignment λ to dampen the executive's power. If R's second

[11] This intuition reflects a similar logic to Alesina and Rosenthal (1996).

period policy π_R is close to 1, the voter could choose $\lambda = 0$ and totally shut down R's ability to pursue $\pi_R = 1$. Naturally, a fully unaligned legislature ($\lambda = 0$) may also pursue policies unaligned with V in the other direction, generating a tradeoff. V thus picks an intermediate value of λ.

$\overline{\delta}$ is also nonmonotonic in b. As b approaches 0, V is more and more willing to stack the legislature against R. As b approaches π_R, the voter and politician are aligned. In both cases, $\overline{\delta}$ approaches ∞. $\overline{\delta}$ achieves a *minimum* at $b = \frac{\pi_R}{2}$, in which case $\overline{\delta} = \frac{4}{\pi_R}$.

Notably, δ can be calibrated numerically using estimates for voters' relative valuations of candidates' horizontal and vertical attributes. Although data on governors is limited, recent structural analysis of Congress by Kawai and Sunada (2025) shows that an extreme congressional candidate who runs against a centrist candidate in a centrist district loses about 4.5 percentage points in vote share (p. 487). Higher valence (or "ability") is associated with a 3.5 percentage point increase. This suggests a δ value of 1.28, far lower than the lower bound of 4 for $\overline{\delta}$ in the theoretical model above. Given governors have greater *managerial* responsibilities than congressional candidates, the value voters place on governors' vertical attributes is likely far greater, meaning their value of δ is likely even lower.

First Period Welfare Finally, I use the above parametrization of voter utility to calculate a measure of *first-period* voter welfare for the incumbent's choice of π^*. Moreover, I also illustrate the optimal degree of legislative alignment λ^* that V would choose, given π^* if he were to choose the composition of the legislature before the first period.[12]

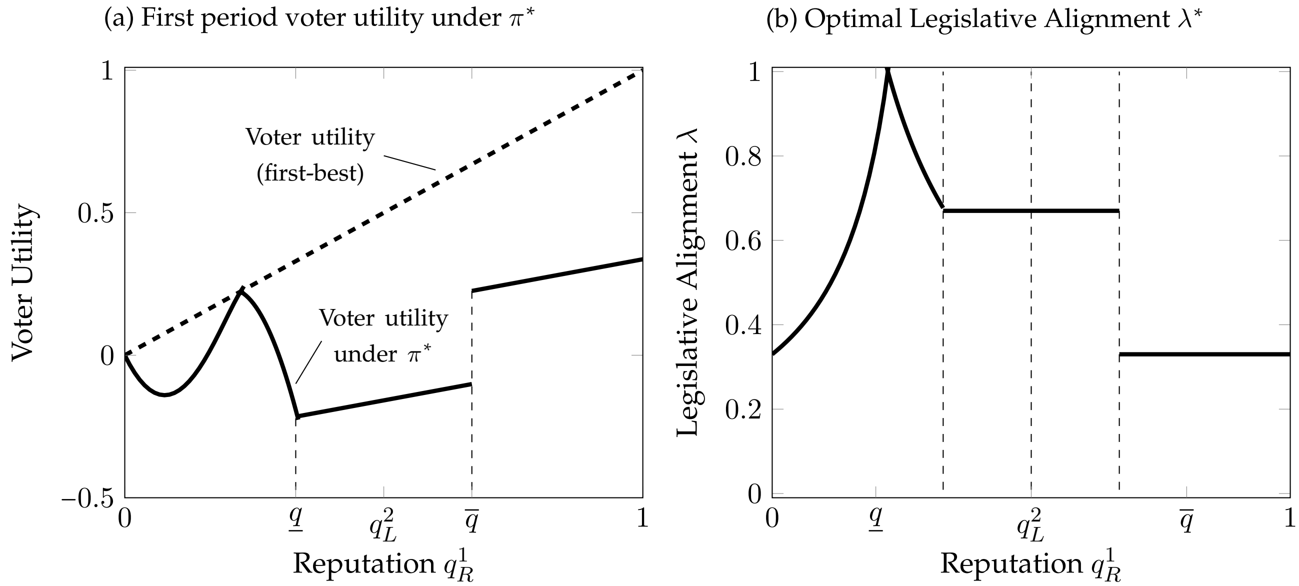
Figure 4 shows voter welfare on the left and the voter's optimal choice of legislative alignment on the right. On the left, voter welfare is benchmarked against the first-best — a politician who would choose $\pi_R = b$, regardless of her reputation, illustrated in the dashed line.[13] For low values of q_R^1, R chooses partisan policies that are unlikely to pass and do not hurt V in expectation. As π^* falls from 1 to b, R's choices become more aligned with V. As π^* falls from b to 0 at $\underline{q}$, policies are less aligned with V. This introduces a nonmonotonicity in welfare. Welfare increases from $\underline{q}$ to $\overline{q}$ as expected ability increases. It counterintuitively jumps at $\overline{q}$ because while $\pi = 1$ is less aligned with V's preferences than $\pi = 0$, it is also easier for R to shut down using the legislature.

The right panel shows how legislative alignment varies with q_R^1 if V could choose legislative alignment before $t = 1$. Alignment increases as R's policy becomes more aligned with V and then falls. V never chooses a legislature fully unaligned with R since it would be pursue policies unaligned with V in the opposite direction..

[12] The figure has a similar shape if λ is fixed.

[13] In cases where $\pi_R < b$, we utilize the symmetry of the proposition above and assume the legislature implements a policy $\lambda(1 - \pi_R)$.

Figure 4: First Period Voter Utility Given Optimal Choice of Legislative Alignment λ^* and R's Optimal Policy π^*



Voter Mobilization Are the predictions of the theory robust to considering the interaction of partisanship with voter turnout? In particular, an incumbent who pursues partisan policies may alienate centrist voters, mobilize opposition voters, or mobilize her own base. Thus, the partisanship of R's policy agenda may affect the threshold ability necessary to win reelection. Formally, suppose R is reelected if and only if $q_2^R \geq q_2^L + z(\pi)$ where $z(\cdot) \in (-1, 1)$ is a function of the agenda's partisanship π. $z(\pi)$ may be increasing in π (partisanship mobilizes R's voter base) or decreasing in π (partisanship alienates voters or provokes the opposition).

As long as $q_2^L + z(\pi) \in (0, 1)$, the results of the model still hold. Intuitively, the nonmonotonicity in Proposition 1 is driven by the convex-concave structure of R's value function — i.e. that she is reelected if and only if her ability surpasses a threshold. Thus, if q_R^1 is low, passing a bipartisan policy may still not be a strong enough signal to be reelected, even if the stakes for reelection are lower. On the other end, if q_R^1 is extremely high, failing to pass a bipartisan policy will still be a damning signal of her ability. Thus, even if pursuing $\pi = 1$ raises the reelection stakes, R would still be better off pursuing this than a bipartisan policy $\pi = 0$.

2.4 Persuasion Model and Data Predictions

While the benchmark in Section 2.2 is useful for conveying the core intuitions of the theory, the information structures it captures are stark. The benchmark does not assume, for instance, that high ability incumbents may be better at passing partisan policies when unaligned with their legislature. Additionally, the incumbent in the model, immediately prior to an election, always wins with probability 0 or 1. This latter point is important for empirical tests, since proxies for winning probability (such as approval ratings) are readily available.

I now describe an extended model utilizing a constrained persuasion structure. This approach gives the incumbent flexibility in directly choosing success probabilities for high and low ability incumbents. The extended model not only shows that the insights of the benchmark are more robust to a general setting, but also allows me to solve the model with a valence shock and thereby write the theoretical predictions in terms of an incumbent's ex-ante winning probability. Additionally, it allows me to easily address two extensions — described in section 2.5 — for a multi-period version of the model as well as an extension regarding politician flow utility.

Persuasion Model I now characterize R's choice of policy agenda as a signal structure $(p_1, p_0) \in [0,1]^2$ for $p_1 \geq p_0$. These respectively represent the conditional probabilities an $a_R = 1$ and 0 incumbent would be able to pass an agenda if she were totally unaligned with the legislature. Total passage probabilities are:

$$a_r = 1: \qquad \lambda + (1-\lambda)p_1, \tag{2}$$

$$a_r = 0: \qquad (1-\lambda)p_0. \tag{3}$$

The partisanship π of an agenda (p_1, p_0) is defined as

$$\pi(p_1, p_0) = 1 - \frac{p_1 + p_0}{2} \tag{4}$$

All else equal, the probability of passage is increasing in ability and decreasing in partisanship.

This model is a constrained persuasion problem. When $\lambda = 0$, the problem boils down to a standard Bayesian persuasion problem with two states (high/low ability) and two signals (success/failure). (p_1, p_0) represent the conditional probabilities of observing a success conditional on each state. For $\lambda > 0$, R chooses a signal structure (p_1, p_0), but the voter V only receives a signal from this structure with probability $1 - \lambda$. With probability λ, he receives a signal from a structure $T = (1, 0)$ (success $\iff a_R = 1$), and cannot differentiate whether the signal they received came from P or T. This constrains the set of posteriors over ability R can induce.

I use the flexibility of this more general model to solve the model with a valence shock. Let $\epsilon \sim N$ be a single-peaked preference shock favoring L, which has support on $\mathbb{R}$, has mean 0, is symmetric around 0, is twice continuously differentiable, and is strictly increasing (decreasing) above (below) 0. V retains L if and only if $q_R^2 \geq q_L^2 + \epsilon$, meaning she wins the election with ex-ante probability $N(q_R^2 - q_L^2)$. Her expected value from an agenda (p_1, p_0) is then

$$\max_{p_1 \geq p_0} \int_{q_R^2} N(q_R^2 - q_L^2) dF(q_R^2 | q_R^1, p_1, p_0).$$

where $F(q_R^2 | q_R^1, p_1, p_0)$ now represents the distribution of posteriors given (p_1, p_0). Let $p^*(q_R^1) = (p_1^*, p_0^*)$ represent the agenda that solves the equation above and π^* the partisanship of this agenda. The generalization of the first proposition is as follows.

Theorem 1. *There exist thresholds $\underline{q} < \overline{q}$ such that:*

- *for $q_R^1 \leq \underline{q}$, $\pi^*(q_R^1)$ is strictly decreasing. $p^*(0) = (1,0)$ and $\pi^*(0) = 1/2$, while $\pi^*(\underline{q}) = 0$ and* $\pi^*(\underline{q}) = 0$.
- *For $q_R^1 \in [\underline{q}, \overline{q})$, $p^* = (1,1)$ and $\pi^* = 0$;*
- *for $q_R^1 \geq \overline{q}$, $p^* = (0,0)$ and $\pi^* = 1$.*

Note in particular that $p^* = (1,1)$ corresponds to the $\pi = 0$ policy in the initial model. The $p^* = (0,0)$ policy corresponds to the $\pi = 1$ policy in the initial model.

As a (constrained) persuasion problem, I utilize insights from the Bayesian Persuasion literature to address parts of the A.1proof. The optimal solution in an unconstrained persuasion problem is given by the concavification of the incumbent's value function. If a policy agenda induces posteriors that achieve the concavification of $N(q_R^2 - q_L^2)$, that agenda is optimal. I show that for $q_R^1 < \underline{q}$, we can indeed use (p_1, p_0) to achieve that concavification, which is a lottery between some posterior $\overline{q}_R^2 = \tilde{q} > q_L^2$ and $\underline{q}_R^2 = 0$.[14] However, because the persuasion problem is *constrained*, the concavification is no longer achievable after some $\underline{q}$. At $\underline{q}$, the solution itself is given by $(p_1, p_0) = (1,1)$. After $\underline{q}$, the incumbent pursues a boundary solution. I show that the solution is either given by $(p_1, p_0) = (1,1)$ or $(p_1, p_0) = (0,0)$, as in the baseline model. I show that if V is an everywhere concave function, then $(p_1, p_0) = (0,0)$ is the optimal solution. Because $N(q_R^2 - q_L^2)$ is locally concave (and equal to its concavification) when q_R^2 is sufficiently high, $(p_1, p_0) = (0,0)$ is the optimal solution for high q_R^2. Otherwise, between $\underline{q}$ and $\overline{q}$, R continues to implement $(p_1, p_0) = (1,1)$.

The persuasion extension suggests a potential *asymmetry* in nonmonotonicity relative to the baseline model. When $q_R^1 = 0$, $\pi = 1/2$, while when q_R^1 is high, $\pi = 1$. The baseline comparative statics also emerge from the proof of the result. Increasing λ causes $\underline{q}$ to decrease and $\overline{q}$ to increase. Increasing q_L^2 causes $\underline{q}$ and $\overline{q}$ to shift to the right.

Data Mapping The expression $N(q_R^2 - q_L^2)$ can be inverted to write the predictions of Theorem 1, as well as the comparative statics, in terms of win probabilities. Let P represent the incumbent's ex-ante probability of reelection.

Corollary 2. *There exist thresholds $\underline{P} < \overline{P}$ such that:*

- *when $P \leq \underline{P}$, partisanship π^* is strictly decreasing in P until $\pi^* = 0$ at $\underline{P}$;*
- *when $P \in (\underline{P}, \overline{P})$, partisanship $\pi^* = 0$;*
- *when $P \geq \overline{P}$, partisanship $\pi^* = 1$.*

Moreover,

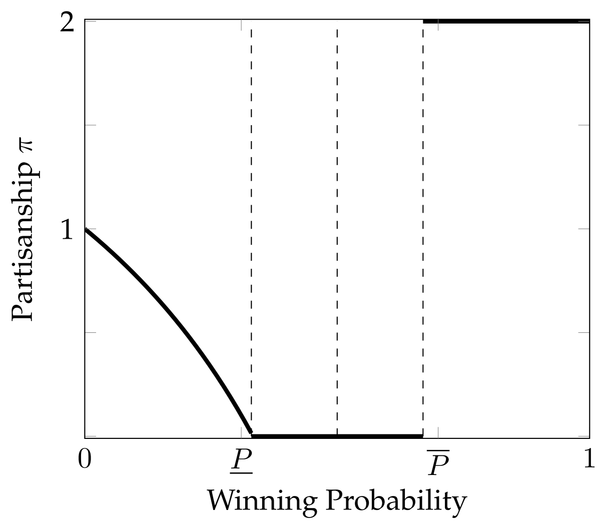
- *When legislative alignment λ increases, $\underline{P}$ decreases and $\overline{P}$ increases.*

[14] A picture of the value function and its concavification is displayed in the appendix.

- *When competition q_L^2 increases, $\underline{P}$ and $\overline{P}$ both increase.*

This corollary summarizes the key predictions I will take to test the partisanship of gubernatorial policy agendas in the model, and is graphed below.

Figure 5: Partisanship of Optimal Policy $\mathbf{P}^*$



The data predictions are as follows:

- **P1**: When looking at the difference in the partisanship of reelectable and term-limited incumbents, there is a nonmonotonic relationship between approval and partisanship. Unpopular incumbents should pursue relatively partisan agendas, moderately popular incumbents bipartisan agendas, and popular incumbents very partisan agendas.

- **P2**: When increasing legislative alignment, the nonmonotonicity of **P1** should flatten out. However, the minimum level of partisanship of **P1** at moderate levels of win probability should also increase.

- **P3**: At high levels of party entrenchment, the nonmonotonicity of **P1** should flatten out; the nonmonotonicity should be strongest in states that lack party entrenchment.

P1 emerges because the model isolates informational incentives for partisan policies that drive reelection. **P2** utilizes the fact that the definition of a bipartisan policy is relative to the legislature's composition, measured using λ. This means that raising λ may also raise the base level of partisanship in the data. **P3** emerges because increasing competition q_L^2 is analogous to increasing party entrenchment.

Finally, although direct measures of winning probability are hard to track historically for executives — particularly in non-election years — approval data are readily available. These data are often used by politicians themselves to assess their popularity and chances at reelection; as such, I utilize approval data to proxy for ex-ante win probability in the data section.

2.5 Extensions

I utilize the persuasion model without a valence shock to address two extensions to the model.

Politician Utility and L Policies First, I show that the model is robust to allowing R to choose policies from the opposing party. In particular, a more partisan policy's informational utility is that it is "hard to pass." However, if R is incredibly aligned with her legislature, policies favored by opposing parties may in fact be harder to pass. Paradoxically, R may attempt to go against her party's preferences and pursue an agenda of the opposition to "prove her worth."

I extend the model by allowing R to choose from a spectrum of partisan policies. She can choose either an R-partisan agenda $p^R \in (p_1^R, p_0^R)$ or an L-partisan agenda $p^L = (p_1^L, p_0^L)$. I endow R with a preference for R policies over L policies. I show that the model's central nonmonotonicity is left untouched. If flow utility concerns are weak, the region in $[\underline{q}, \overline{q}]$ may exhibit additional nonmonotonicities in the direction of L policies.

Suppose that R chooses a policy agenda p^R or p^L. Success probabilities are:

$$\text{Choose } \pi_R : \lambda a_R + (1-\lambda) p_{a_R}^R$$
$$\text{Choose } \pi_L : (1-\lambda) a_R + \lambda p_{a_R}^L$$

If R chooses π^L, she is adopting the policies an L politician would implement, meaning her de facto legislative alignment flips. I assume R receives a disutility $-c$ from pursuing π^L.

Let $F^i(q_R^2 | q_R^1, p_1^i, p_0^i)$ be the distribution of posteriors over R's ability for $i \in \{R, L\}$. R implements an agenda from π^R if and only if

$$\max_{p_1^R \geq p_0^R} \int_{q_R^2} u_R(q_R^2) dF^R(q_R^2 | q_R^1, p_1^R, p_0^R) \geq -c + \max_{p_1^R \geq p_0^R} \int_{q_R^2} u_R(q_R^2) dF^L(q_R^2 | q_R^1, p_1^L, p_0^L).$$

The left expression is the value from the solution to the baseline problem. The right expression is the value from choosing an L policy agenda less the cost $-c$ from choosing policies preferred by the opposing party. We can write the optimal solution of each program as p^{i*}, which is characterized by thresholds $\underline{q}^i, \overline{q}^i$. While c sufficiently large may mechanically shut down any preference for L policies, I show that even for c small, the model's nonmonotonicity still holds.

Proposition 3. *There exists $\overline{\lambda} > 1/2$ such that for $\lambda \leq \overline{\lambda}$, R implements p^{R*}. For $\lambda \geq \overline{\lambda}$ and c sufficiently low, there exist thresholds $\underline{q}^f, \overline{q}^f$, with $\underline{q}^R < \underline{q}^f < \overline{q}^f < \overline{q}^R$such that R implements*

- *p^{R*} for $q_R^1 \leq \underline{q}^f$;*
- *p^{L*} or p^{R*} for $q_R^1 \in (\underline{q}^f, \overline{q}^f)$;*
- *p^{L*} for $q_R^1 \geq \overline{q}^f$;*

The intuition is as follows. Below $\underline{q}^R$ and above $\overline{q}^R$, R is able to achieve the concavification of $u_R(\cdot)$ utilizing p^R. The integrand term on the left hand side is maximized, suggesting a strict preference for p^{R*}. Note that R's maximal utility from pursuing p^{R*} is strictly decreasing in λ — since lower λ allows greater informational flexibility — while the maximal utility from pursuing p^{L*} is strictly increasing in λ. The utility from pursuing either policies is always equal at $\lambda = 1/2$, modulo the $-c$ term. This means that p^{R*} is still strictly preferred at $\lambda = 1/2$. Preference for p^{L*} can override if and only if c is sufficiently small; and only then when the value from pursuing π^{L*} is sufficiently higher than that of p^{R*}. This can occur only within a strict subset of $[\underline{q}^R, \overline{q}^R]$.

Extending the space of partisanship downwards — so that L policies are even *less* partisan than $\pi = 0$ (for an R incumbent) — suggests that partisanship weakly decreases from $\pi = 1/2$ to 0 at $\underline{q}^f$; potentially plummets further between $\underline{q}^f$ and $\overline{q}^f$, either with the implementation of a moderate or extremely partisan L policy; and then shoots back up to $\pi = 1$ above $\overline{q}^f$. That is, within $(\underline{q}^f, \overline{q}^f)$, R implements either $\pi^R = 0$ or $\pi^L \geq 0$, potentially adding additional nonmonotonicities to the pattern, but ensuring that partisanship (relative to R's party) is always lower in this middle region than when reputation is low or high.

Multi-Period Dynamics I show that allowing for an additional period of policy passage preserves the fundamental insights of the model. While the baseline model assumes that politicians' policy agendas represent the totality of their term in office, we can consider a three period model where politicians pursue agendas in periods $t = 0$ and $t = 1$ before an election at $t = 2$. In between $t = 0$ and $t = 1$, agents update beliefs over a_R as before.

Following notation from earlier, let the belief that R is high ability at the beginning of $t = 0$ be q_R^0. The A.1optimal solution is characterized as follows.

Proposition 4. *Given $\underline{q}, \overline{q}$, there exist thresholds $\underline{\underline{q}} < \underline{q} < \overline{q} < \overline{\overline{q}}$ such that*

- *For $q_R^0 \leq \underline{\underline{q}}$, partisanship π^* is decreasing from $1/2$ at q_R^0 to 0 at $\underline{\underline{q}}$.*
- *There exists $q_\dagger \in [\underline{\underline{q}}, \overline{q} < \overline{\overline{q}}]$ such that for all $q_R^0 \geq q_\dagger$, $\pi^* < 1$.*
- *For $q_R^0 \geq \overline{\overline{q}}$, $\pi^* = 1$.*

3 Data

This section begins with an overview of U.S. gubernatorial State of the State addresses and voter assessments of governors vis-á-vis their policy agendas. I then discuss how the data is sourced and pre-processed before analysis. I finish by detailing how I use large-language modeling to isolate policy proposals from speeches.

3.1 U.S. Governors and State of the State Addresses

Reputational Priorities of U.S. Governors The policy agendas of U.S. governors provide an ideal setting to test the predictions of the model. Governors are high-profile political executives who, since the end of the 20th century, have played an outsized role in setting, pursuing, and implementing state policy agendas. Governors "enjoy organizational, institutional, and popular advantages similar to and arguably even greater than the president [over Congress]" (Heidbreder, 2012). In any given year the 50 U.S. governors in office all vary in their popularity, eligibility for reelection, party entrenchment, and alignment with state legislatures.

There is substantial evidence that governors are evaluated based on their productivity in passing their policy agendas. A guide provided by the National Governors Association (NGA) to incoming governors explicitly remarks that "[t]he media and public will judge the governor's leadership ability and success... [by] whether the administration's legislative program succeeds... The governor's ability to manage and secure legislation also affects his or her ability to serve as a strong leader of the party... If the legislation fails, it will be considered a political defeat... [T]he passage of priority legislation usually will signal a political success." (National Governors Association, 2018). An executive director of the NGA has gone as far as to say that "the ultimate measure of success [is] the ability of... governors to get his or her initiatives enacted" (Scheppach, 2005).

The NGA guides also provide insights on the dimensions of skill by which a governor is evaluated, often using the language of a managerial "CEO" to describe these responsibilities. "As chief executive officers (CEOs), governors are responsible for the leadership and management of their states. As leaders, they set priorities for their administration and enact new policies and programs designed to achieve those priorities" (National Governors Association, 2019). Moreover, an "effective process to craft and implement a legislative program and strategy, as well as to cultivate and maintain working relationships with legislative leaders and members, is critical to ensuring the success of a governor's legislative program" (National Governors Association, 2018).[15]

A long literature details how voters reward political executives for performance in office, including agenda passage. Besley and Case (1995), Wolfers et al. (2002), and Wolak and Parinandi (2022) find that voters reward governors for (state-specific) economic outcomes and policy outputs. Policy successes boost the share of seats a governor's party holds in the legislature (Lebo & O'Geen, 2011). In line with the theory, voter responsiveness to executive performance is also mediated by an executive's alignment with the legislature. Voters are more likely to blame opposing parties for problems if the opposition holds an executive position and their own party holds the legislature (Brown, 2010). Voters are much more responsive to changes in state economic outcomes when the governor and legislature are from the same party (Leyden & Borrelli, 1995). Larimer (2015) argues that "[u]nified control provides the public an easy target to blame when things go bad" (p. 96), going on to note that divided governments may permit flexibility in allowing voters to learn about governors' managerial strengths.

[15] Anecdotal evidence supports the idea that executives strengthen their reelection chances by successfully pursuing ambitious policies in the face of partisan adversity. Iowa Governor Terry Branstad, for instance, describes how his ability to massively restructure state government aided in a competitive reelection to his second term. Branstad writes that he managed to secure "90 percent of the reorganization package" despite a "legislature overwhelmingly controlled by the opposition party" (Behn, 1991), going on to describe the techniques of solid management, legislative persuasion, and drive necessary to achieve those goals.

Policy Agendas The model's predictions apply to executives' policy agendas. I utilize the text of each governor's annual "State of the State" address as a baseline corpus of policy agendas. At the beginning of each legislative session, the governor of each state in the United States is required to address a joint session of the legislature to deliver a "State of the State" (SotS) speech. [16] The speech is delivered annually (in some states biennially) in the first quarter of the year, and is analogous to the presidential "State of the Union." Since these speeches are given around the same time every year in (almost) every US state, these data allow an annual panel of policy agendas for each governor, state, and year.

According to a National Governors Association guide on legislative relations, the "inaugural address, State of the State address and budget message are all excellent forums to communicate and build momentum for the executive branch's legislative agenda" (National Governors Association, 2018). A large literature in political science has documented the importance of these speeches as vehicles for the governor reflect on her administration's past accomplishments and lay out her policy priorities for the coming year (Rosenthal, 2012). Coffey (2005) and Heidbreder (2012) argue that these speeches accurately represent the incumbent's current priorities rather than pure policy preferences. The agenda items the governor pursues are picked strategically, anticipating "the reaction of the legislature, based on their political intelligence and on their consultations with legislative leaders and other members" (Rosenthal, 2012).

Governors pass many of the agenda items laid out in their addresses, suggesting that these speeches are not purely "cheap talk." Kousser and Phillips (2012) investigate over $1,000$ proposals from State of the State speeches in the mid-2000s, showing that 41% passed in relatively unadulterated form and 18% with compromises. One implication is that there are basic duties (such as budget management) that a skilled executive should be able to address. An ex-governor of Maryland noted that "[g]overnors must ensure that the budget is balanced, that the state can adequately respond to its day-to-day challenges, and must be able to work with lawmakers from both parties and across the ideological spectrum" (Kousser & Phillips, 2012, p. 97).

Comprehensive analysis of partisanship in State of the State speeches has been scarce beyond a few years worth of data (Coffey, 2005; DiLeo, 1997; Heidbreder, 2012; Warner, 2023; Weinberg, 2010). DiLeo (1997) shows governors in Democratic states are more likely to pursue redistribution. Ferguson (2003) shows that governors' priorities are disciplined by legislative compositions and economic conditions. Coffey (2005), Weinberg (2010), and Kousser and Phillips (2012) measure ideology by manually coding sentences or relying on dictionaries of words. Heidbreder (2012) shows that Democratic governors are more attentive to healthcare and social policies.

One of the key issues with analysis of SotS addresses is the lack of a systematic, centralized data source of speeches, with the exception of two potential databases. Lushkov (2019) collects hundreds of SotS addresses back to the 1800s to look at the frequency with which governors discuss education. Butler and Sutherland (2023) have digitized almost all SotS addresses from 1960

[16] In some states, the speech is called the "State of the Commonwealth"; in others, the governor's budget or inaugural address take the same role.

onwards, documenting an increase in nationalization of speech. Hopkins et al. (2022) document a divergence in the speech of 1,783 state party platforms from 1918-2017 beginning in the 1990s, but restrict their analysis to the frequencies of certain topics and phrases.

3.2 Sourcing and Pre-Processing

I compile a digitized text library of the State of the State speeches from 1990-2020, allowing me to employ methods from the literature on partisanship of speech and large-language-modeling to measure partisanship, study its heterogeneity, and test the model's predictions.

I began with the database of Lushkov (2019), which contained 330 speeches post-1995 due to data loss. Next, I scoured educational resources, archived governor websites, and state libraries to assemble SotS speeches for an initial dataset of $1,144$ usable speeches. I later gained access to data from Butler and Sutherland (2023), allowing me to assemble a final dataset of $1,345$ speeches for 1990-2020. Figure B.1 in the appendix graphs the set of usable speeches. I broke these speeches into snippets of thematically contiguous thoughts using the NLTK (Natural Language Toolkit) Text-Tiling tokenizer. Each snippet was about 7 sentences and can be thought of either as a paragraph. I end up with $78,702$ snippets for the period $1990-2000$.

3.3 Speech Processing with BERT

To separate portions of SotS addresses discussing the governor's policy agenda, I fine-tuned a "BERT" model from the huggingface transformers library to identify relevant snippets of text. I asked it to classify whether a given snippet of text discussed a concrete policy (yes/no) and, if so, whether it was a discussion of a policy proposal or made mention of a past policy (yes/no).

BERT is a pretrained model that learns the structure of provided text examples, allowing it to be "fine-tuned" to classify text (Devlin et al., 2018). I use the same methodology as Card et al. (2022), which identifies whether Congressional speeches were about immigration and, if so, their tone (positive/negative/neutral). I repurpose much of these authors' github code to my setting.

Hand-Coding I randomly selected around 9500 snippets from the $70,368$ snippets of the initial $1,144$ speech 1995-2020 dataset to be hand-labelled and provided to BERT as fine tuning data. Two research assistants were given the following instructions to code these snippets.

1. "Policy." Coded as "1" if the snippet discusses the enactment of a state-level policy (either passed by the governor, state government, or referendum) and "0" if it does not. A policy discussion is a reference to a specific act of legislation or law, a concrete proposal to increase or decrease funding to a certain cause, other legal orders proposed by the government to take certain concrete actions, and discussions of details of any of the above.

2. "Proposal/Past." Only applies if "policy" coded as "1." Coded as "1" if the snippet refers to a policy that has just put into place or will be put into place in the future. Coded as "0" otherwise — in particular, if a governor is reflecting on the effects of a policy in the past.

The full coding guide is contained in Appendix C. Of these hand-coded data, we identified $4,144$ snippets as discussing policy and, of those, $2,993$ as discussing proposals.

Detailing BERT Model I utilized the "BERT-Base-Uncased" model, which contains approximately 110 million parameters which are adjusted through the process of fine-tuning, to analyze the data. I make use of a two-layered model approach. I first train a model to classify whether a snippet is a policy discussion, which takes in as inputs all the snippets in our hand-coded data. The second model is trained to classify whether a snippet is about a policy proposal or some reflection on a past policy.[17] Once these models were fine-tuned, I ran the full dataset of snippets through each model. I used the policy classification to identify policies, and then the proposal classification model to identify proposals, allowing me to classify each snippet as desired: snippets about policy proposals, snippets about policy but not proposals, and other snippets. I additionally hand-coded a held-out test set of 600 snippets to assess model performance. The policy classification model ended up with a mean-squared-error 0.141 (meaning the classifier agreed with the test set 86% of the time). The mean-squared-error for the combination of the policy and proposal classifiers was 0.136. Of the $78,702$ snippets, I identify $35,636$ (45%) as corresponding to policy and $26,720$ (34%) as corresponding to policy proposals.

3.4 Additional Data Sources

The following additional data sources are used in the analysis:

- Governor names/dates in office: National Governors Association (2024)
- Gubernatorial win margins and seat status: Algara and Amlani (2021)
- Quarterly governor approval ratings: Singer (2023)
- Legislative composition: Klarner (2013) and National Conference of State Legislatures (2024)
- Term limit rules and missing election dates: Ballotpedia (2024a) and Ballotpedia (2024b)
- Additional covariates: Grossmann, Jordan, and McCrain (2021), MIT Election Lab (2024), NASBO (2025), U.S. Bureau of Labor Statistics (2026)

4 Documenting Partisanship

The section begins by detailing measurement of aggregate partisanship. I then document aggregate changes in partisanship of policy proposals from 1990-2000. Next, I show that the language

[17]The input (hand-coded) data for each model is divided into 10 folders for 10-fold cross-validation. Each folder takes the hand-coded data and partitions it into three groups. The first set, the "train set," is the main dataset used to train the model. The "dev set," comprised of 400 snippets, is used to adjust the model after its initial training. Finally, the "test set," comprised of 300 snippets, is used to calculate the models' accuracy. The model is first trained using these 10 folders and then run a final time on all the data. Before feeding the data into the model, we also tokenize the data using spaCy, which takes each snippet and breaks it into its component words.

of partisan policy proposals I identify in governors' speeches are indeed associated with language related to policy proposals, and that the most and least partisan language in these proposals are largely associated with partisan policies traditionally associated with Democrats and Republicans. Finally, I describe how aggregate trends disguise substantive variation at the level of governors, states, and regions. I use these insights to motivate how empirical variation is interpretable within my model, which is explored in Section 5.

4.1 Calculating Partisanship

With the corpus of governor speeches, partisanship of gubernatorial speech can be measured for both full speeches and just policy proposals using techniques from the literature on political speech. I calculate a measure of partisanship π_{it} for each governor i and year t. I then take a weighted sum of π_{it} to construct an annual measure of aggregate partisanship Π_t.

The metric for defining and calculating partisanship is identical to Gentzkow, Shapiro, and Taddy (2019). Aggregate partisanship for year t, Π_t, is measured as the expected *informativeness* of a randomly selected phrase in inferring a governor's party. Suppose we randomly select a governor — selecting from one of Democrats or Republicans with (prior) probability $1/2$. Partisanship is measured as the expected posterior of guessing that governor's party correctly. If partisanship is less than or equal to $1/2$, speech is — on average — uninformative of governor party. If Republican and Democratic governors use dissimilar phrases, partisanship is greater than $1/2$.

I use "bigrams" (two-word phrases) as my measure of phrase. I use the NLTK PorterStemmer to reduce words to their base form. I then use the Gentzkow, Shapiro, and Taddy (2019) "leave-out-estimator," for calculating aggregate partisanship Π_t. For a sample period of interest, let R be a set of Republican governors and D Democratic governors.[18] Let c_{ij} be the count of phrase j used by governor i. Let $C_{ij} = \frac{c_i j}{\sum_{j \in J} c_{ij}}$ be the normalized count of phrase j used by governor i, where J is the set of all phrases used in the time period of interest. Let C_j^P be the normalized count of phrase j used by party P: $C_j^P = \frac{\sum_{i \in P} c_{ij}}{\sum_{i \in P} \sum_{j \in J} c_{ij}}$. Finally, let $T(t)$ be a five-year window around time t: $T(t) = \{t-2, t-1 \ldots, t+2\}$.[19] A subscript t represents the value of the variable at time t, while $t \in T(t)$ represents its value within the five-year window. For a phrase j and speaker i, define ρ_{-ijt} as the ratio of j's use by Republican vs. Democratic governors, modulo governor $i \in R_t \cup D_t$, within the five-year window acting as the reference group:

$$\rho_{-ijt} = \frac{C_{jt \in T(t)}^{R-\{i\}}}{C_{jt \in T(t)}^{R-\{i\}} + C_{jt \in T(t)}^{D-\{i\}}}.$$

Concretely, given a flat prior, ρ_{-ijt} is the posterior probability assigned to a speaker being a Re-

[18]The few independent governors in our sample lean Democrat, so I classify them as Democratic for all intents and purposes.

[19]I make use of a five-year window since each year only has at most 50 speeches. Gentzkow, Shapiro, and Taddy (2019) use Congressional Record Data, where the number of speakers and quantity of text is much larger, and the length of a Congressional session is two years.

publican governor upon observing phrase j.

Aggregate partisanship in the US States at time t, Π_t, can then be calculated as:

$$\Pi_t = \frac{1}{2}\frac{1}{|R_t|}\sum_{i\in R_t}\sum_j C_{ijt}\cdot\rho_{-ijt} + \frac{1}{2}\frac{1}{|D_t|}\sum_{i\in D_t}\sum_j C_{ijt}\cdot(1-\rho_{-ijt}). \tag{5}$$

The interpretation of Π_t is precisely as above. With probability $1/2$, we randomly select a party (R or D), and from there randomly draw a governor. With probability C_{ijt}, the chosen governor i uses phrase j. Then, conditional on phrase j, the posterior over governor party moves to ρ_{ijt}. Π_t averages this posterior across governors and phrases.

I measure the partisanship of the speech of governor i in state s at time t as:

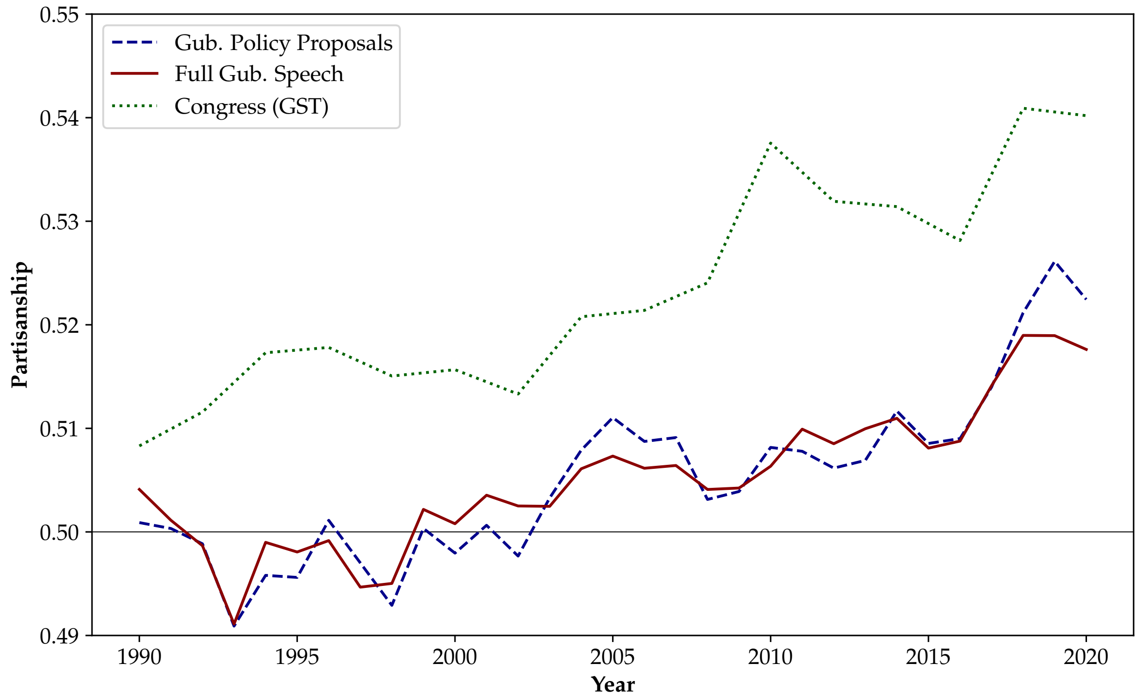
$$\begin{aligned} \pi_{it} &= \sum_j C_{ijt}\cdot\rho_{-ijt} && i\in R_t \\ \pi_{it} &= \sum_j C_{ijt}\cdot(1-\rho_{-ijt}) && i\in D_t, \end{aligned} \tag{6}$$

which measures how partisan governor i's speech is relative to her party. Governors with $\pi_{it}\geq 1/2$ use language mostly in line with their party, while those with $\pi_{it}\leq 1/2$ use language in line with that of the other party.

4.2 Aggregate Behavior of Partisanship

The figure below shows the evolution of partisanship over the sample period for three series. Partisanship calculated using the entire corpus of speeches is shown in the solid red line, while partisanship calculated using only the policy proposal component of speeches is shown in the dashed blue line. Finally, the dotted green line plots the series of Congressional partisanship from Figure 2A of Gentzkow, Shapiro, and Taddy (2019), which also uses the leave-out estimator. In Appendix Figure B.2, contains measures of partisanship for past policy discussions and rhetoric.

Figure 6: Partisanship of U.S. Gubernatorial Speeches, Full Speech vs. Policy Proposals vs. Congressional Record: 1990-2020



Partisanship of gubernatorial speech calculated using Gentzkow, Shapiro, and Taddy (2019) leave-out estimator. "Gub. Policy Proposals" computes estimator for gubernatorial speech snippets coded as discussing policy proposals on corpus of U.S. governors' speech from 1990-2020. "Full Gub. Speech" computes estimator for all gubernatorial speech in given year. "Congress (GST)" series is partisanship of Congressional speech using leave-out estimator, replicating Figure 2A of Gentzkow, Shapiro, and Taddy (2019).

State-level partisanship for full speeches and the policy proposal data remains low in the early 1990s and is essentially negligble until after the year 2000. The policy proposals data do not experience their first prominent increase until a hump lasting from 2001-2008. Partisanship dips during the Great Recession before resuming a slow upward creep. Partisanship spikes in 2017 after the 2016 electoral cycle.

The measure of gubernatorial partisanship appears to track many features of the Congressional partisanship series from Gentzkow, Shapiro, and Taddy (2019), with a few notable exceptions. First, the Congressional series exhibits a spike from 2008-2010 that is absent from the gubernatorial series. Second, the timing of increases in the gubernatorial series is more gradual. Third, the level of gubernatorial speech is lower than that of the Congressional record. In fact, it is not until 2015 that the *level* of gubernatorial partisanship even reaches that of Congressional Partisanship in 1990. Nevertheless, both series experience increases after 2000 that finish with spikes in 2017.

Features of Partisan Proposals While the BERT classifier captures the partisanship of language, does this partisan language actually correspond to concrete policies that are more/less partisan? I.e., is it the case that more partisan policy proposal speech is associated with actual partisan policies, or do governors dress up non-partisan proposals in partisan language?

As a validation exercise, I calculate the most partisan bigrams present in the policy proposals text. I utilize a method from Gentzkow, Shapiro, and Taddy (2019), where the partisanship of a phrase is determined as measuring the informational loss in inferring governor's party when removing phrase j from governors' vocabulary. This formula for phrase j is given by

$$1/2 - 1/2 \sum_{j \neq k} \left(\frac{C_k^R}{1 - C_j^R} + \frac{C_k^D}{1 - C_j^D} \right) \frac{C_k^R}{C_k^R + C_k^D}, \tag{7}$$

where more positive numbers correspond to more Republican phrases and more negative numbers to more Democratic phrases.[20] I calculate this metric for each bigram in the six epochs of 1990-1994, 1995-1999, 2000-2004, 2005-2010, 2010-2014, and 2015-2020.

Table 1 lists the ten "most Republican" and "most Democratic" phrases from the policy proposal text. The figure replicates Table 1 in Gentzkow, Shapiro, and Taddy (2019). It shows that the "most partisan" language is overwhelmingly associated not only with policy proposals, but also policies associated with each party. Republicans are likely to mention taxes — including phrases like tax reduction, tax relief, and cutting taxes. They also emphasize charter schools, the budget and, later in the sample, discuss crime and law enforcement. Democrats are more likely to mention topics related to the environment, welfare and childcare, healthcare and health insurance, the minimum wage, and affordable housing. This suggests that the combination of the BERT classifier with partisanship of political speech delivers concrete measures of partisanship of *policies*.

[20]Following the norm established by their paper, I code Republican governors' party as +1 and Democratic governors' as −1.

Table 1: Most Republican and Most Democratic Phrases, 1990-2020

(a) 1990-1994

Republican	Democratic
properti tax	state govern
gener assembl	long term
sale tax	commun colleg
incom tax	health insur
amend section	human resourc
public school	welfar recipi
act appropri	clean air
tax relief	child care
school district	econom develop
tax reduct	state employe

(b) 1995-1999

Republican	Democratic
properti tax	class size
school district	high school
tax relief	child care
million dollar	public safeti
cut tax	perman fund
charter school	tax credit
budget recommend	transport system
econom develop	privat sector
look forward	year thi
thi budget	econom growth

(c) 2000-2004

Republican	Democratic
charter school	health care
incom tax	prescript drug
high tech	state agenc
mental health	properti tax
center excel	minimum wage
tax cut	domest violenc
million state	school construct
long term	billion dollar
tax relief	health insur
low incom	educ lotteri

(d) 2005-2009

Republican	Democratic
incom tax	health care
tax relief	thi budget
charter school	health insur
million dollar	new job
math scienc	feder govern
tax rate	clean energi
gener fund	energi effici
qualiti life	pre k
properti tax	creat job
budget provid	afford health

(e) 2010-2014

Republican	Democratic
incom tax	health care
econom develop	sale tax
charter school	tax credit
state govern	creat job
budget recommend	minimum wage
school district	earli childhood
feder govern	mental health
job creator	21st centuri
high school	gener assembl
gener fund	thi budget

(f) 2015-2020

Republican	Democratic
incom tax	clean energi
tax relief	afford hous
budget recommend	health care
high school	minimum wage
law enforc	middl class
tax cut	renew energi
properti tax	climat chang
task forc	child care
pay rais	let pass
depart correct	work togeth

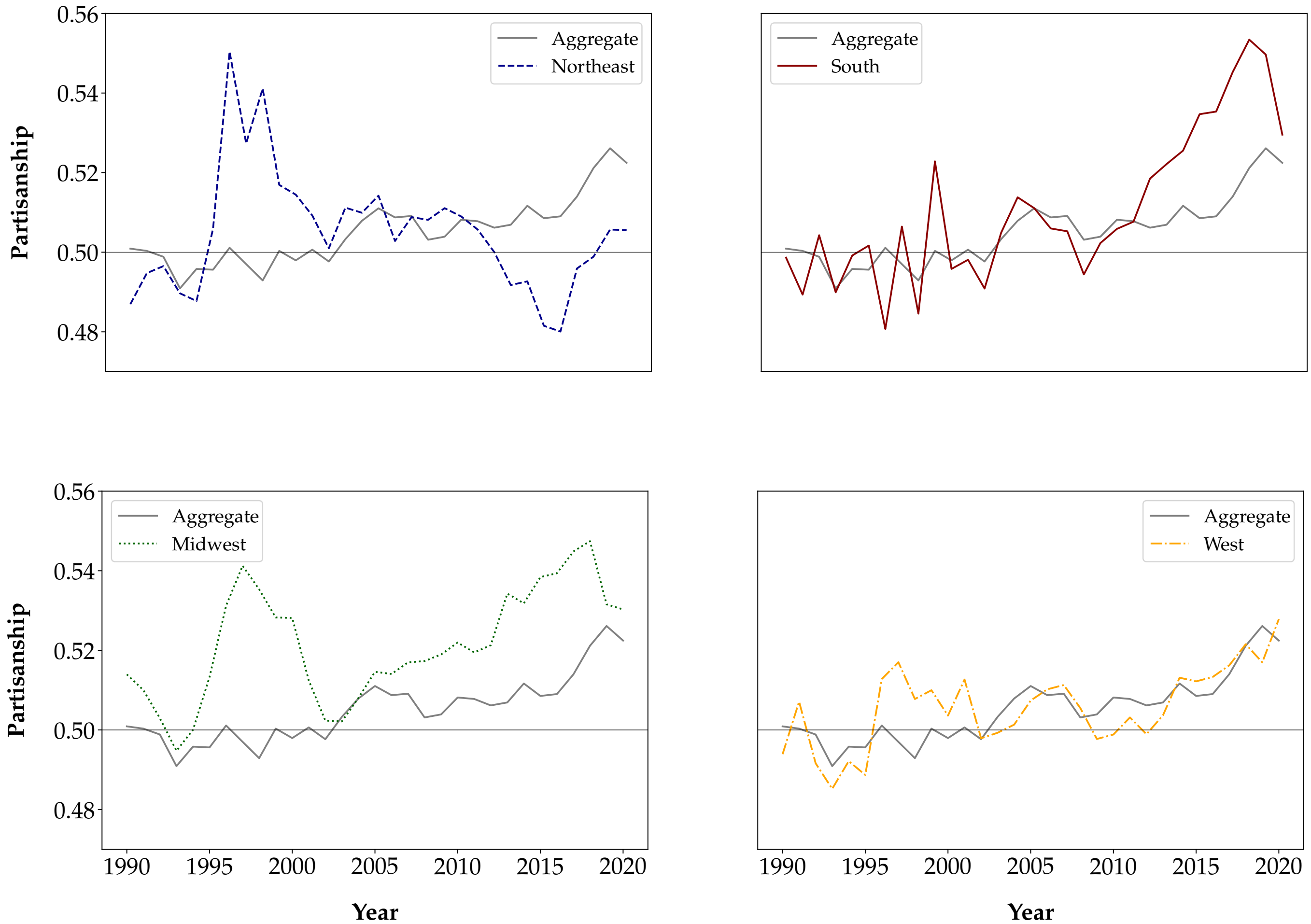
Table lists top 10 most Democratic and top 10 most Republican bigrams, in descending order, for each time period. Partisanship of phrase measured using equation (7), which describes informational loss in inferring governor's party based upon removal of bigram, as in Gentzkow, Shapiro, and Taddy (2019) Table 1. Table removes certain procedural phrases in calculation of partisanship of phrase use.

4.3 Heterogeneity

The aggregate series in Figure 6 mask substantial heterogeneity in partisanship at the state level. Taking the policy proposal data, the figure below shows this heterogeneity by displaying the average of governor-level partisanship π_{it} for each of the four U.S. Census regions.

Figure 7: Partisanship of U.S. Gubernatorial Policy Proposals, by U.S. Region: 1990-2020

(a) Northeast (b) South



(c) Midwest (d) West

Partisanship of gubernatorial speech calculated using Gentzkow, Shapiro, and Taddy (2019) leave-out estimator on U.S. gubernatorial speech snippets coded as discussing policy proposals. Northeastern states are CT, ME, MA, NH, NJ, NY, PA, RI, VT. Midwestern states are IL, IN, IA, KS, MI, MN, MO, NE, ND, OH, SD, WI. Southern states are AL, AR, DE, FL, GA, KY, LA, MD, MS, NC, OK, SC, TN, TX, VA, WV. Western states are AK, AZ, CA, CO, HI, ID, MT, NM, NV, OR, UT, WA, WY.

The Northeast and Midwest both experience spikes in partisanship at the end of the 1990s, well-surpassing the apex of partisanship achieved at the end of the aggregate series. Partisanship in the Northeastern states then experiences a trough around 2015 and ends at the same level it began in the 1990s The Southern states in panel (b) exhibit significant variation in partisanship before 2005. The average level of partisanship for these states accelerates throughout the 2010s but then dips after 2017. The Western States in panel (d) follow the aggregate trend most closely, but even these states see regular dips and troughs. This regional heterogeneity is indicative of deeper variations in partisanship at the state and governor level. Table B.1 in the appendix cursorily shows that there is meaningful variation in the partisanship of speech within states and governors. The next section utilizes the insights of the theoretical model to investigate how differences in partisanship can emerge from differences in electoral incentives.

5 Testing the Model

This section tests the predictions of the theoretical model on the data for US governors. I first show that many of the key assumptions of the model are borne out in proposal-level data. I use data from Kousser and Phillips (2012), who manually extracted policy proposals from State of the State speeches and tracked their progress. As the theory predicts, a one standard deviation increase in agenda partisanship means proposals from that agenda are up to *nine* percentage points less likely to pass. Additionally, high approval governors are more likely to pass their agendas, and this is primarily driven by governors aligned with their legislatures.

Next I present evidence supporting the model's behavioral predictions using my data on partisanship. I show that the U-shape presented by the theory can rationalize empirical variation in partisanship with governor approval and legislative alignment. Afterwards, I disaggregate the results by degree of electoral competition and state type to show how further empirical findings can be rationalized by the theory's comparative statics.

To match the fundamental variables present in the theory — partisanship, win probability, and legislative alignment — I use the following notation.

- Partisanship, π_{ist}. I use the partisanship formula in equation (6) and normalize it by the sample mean and standard deviation.[21] I use the policy *proposal* measure of partisanship.

- Win probability, q_{ist}. I use quarterly gubernatorial approval data compiled by Singer (2023).[22] Since SotS addresses are given in Q1 of each year, I use approval data from Q4 of the previous year. For governors' inaugural years, I use the approval data from Q1.[23]

- Legislative alignment, λ_{ist}. I combine the data on legislative composition from Klarner (2013) and National Conference of State Legislatures (2024). I choose a binary representation of λ_{ist}. The variable is equal to 1 if more than 50% are members of the governor's party and 0 otherwise. This variable combines both legislative chambers.

[21] The sample mean is 0.511 and the standard deviation 0.042.

[22] I assume win probability is a strictly increasing function of gubernatorial approval within each state.

[23] For the sample period, these data are missing from Idaho. There are also substantial gaps for Hawaii, Louisiana, and North Dakota.

5.1 Partisanship and Policy Passage

Policy Passage and Partisanship in 2001 and 2006 The key assumption underlying the theory is that partisan agendas are harder to pass. To test this, I utilize replication data from Kousser and Phillips (2012), who hand-collected State of the State addresses for a subset of states for 2001 and 2006, manually identified proposals in these speeches, tracked any corresponding bills in the legislature, and then coded whether that proposal passed (e.g. as a legislative bill or an executive order) or was a compromise. [24] These data have the advantage that they are an exhaustive categorization of the outcome of any and all policies proposed by governors in these years.

Let $pass_{jist}$ be an indicator equal to one if a policy j from a governor i's agenda passes in state s at time t. This variable is coded as 0.5 by Kousser and Phillips (2012) if a bill passed as a "compromise" relative to a governor's proposal. To investigate the relationship between policy passage and partisanship, I run variants of the following regression.

$$pass_{jist} = \theta_0 + \theta_1 \pi_{ist} + \theta_2 q_{ist} + \theta_3 \lambda_{ist} + \theta_4 \lambda_{ist} \cdot q_{ist} + \chi_t + \zeta_{ist}. \tag{8}$$

χ_t is a year fixed effect. Equation 4) of the theory predicts that passage should be negatively correlated with partisanship ($\theta_1 < 0$). It also predicts that θ_2 is weakly positive (policy passage weakly increases with approval for governors with unaligned legislatures), while $\theta_2 + \theta_4$ is positive (policy passage strictly increases with approval for governors with aligned legislatures).[25]

Table 2 first regresses policy passage only on partisanship and then gradually adds the other covariates in equation (8). It shows a negative and significant relationship between policy passage and partisanship of agendas. Column (1) shows that, without including any additional covariates, a one standard deviation increase in partisanship means a proposal is 3.5 percentage points less likely to pass. The effect is much larger in column (2) at 6.5 percentage points. Approval by itself does not substantively affect passage, but an aligned legislature does.

[24] I would like to thank Justin Phillips for providing the replication data from their book.

[25] Note that the general formulation of the model in equation (4) does not suggest a clear relationship between legislative alignment and partisanship; rather, all else equal, policies should be less likely to pass the more partisan they are.

Table 2: Regressions of Policy Passage from Kousser and Phillips (2012) on Partisanship and Other Covariates, 2001 and 2006

	(1)	(2)	(3)	(4)
Partisanship	-0.034^{**}	-0.066^{***}	-0.069^{***}	-0.089^{***}
	(0.0173)	(0.0181)	(0.0180)	(0.0241)
Approval		0.243	0.0864	0.131
		(0.198)	(0.314)	(0.422)
Leg. Align		0.116^{***}	−0.111	−0.305
		(0.0295)	(0.220)	(0.299)
Leg. Align × Approval			0.478	0.908
			(0.402)	(0.551)
Constant	0.497^{***}	0.294^{***}	0.300^{*}	0.238
	(0.0140)	(0.111)	(0.176)	(0.236)
Approval + Leg. Align × Approval			0.564^{***}	1.039^{**}
			(0.255)	(0.355)
Mean of Dep. Var	0.500	0.496	0.496	0.472
N	1053	949	949	545
Num. States	28	25	25	25
Year F.E.	No	No	Yes	Yes
Legislative Only	No	No	No	Yes
R^2	0.004	0.027	0.045	0.064

Regression of policy proposal passage on selected covariates. Dependent variable is binary variable equal to one if and only a policy proposal is passed, taken from Kousser and Phillips (2012). Bills passed as "compromises" relative to governors' proposals coded as 0.5. Dependent Partisanship of U.S. governor State of the State speeches measured at the state-year level using leave-out estimator of Gentzkow, Shapiro, and Taddy (2019), as calculated in equation (6), normalized by sample mean and standard deviation. Governor approval utilizes previous year's fourth quarter approval rating from Singer (2023) and current year's first quarter rating if unavailable. Legislative alignment is binary and equal to one if and only if more than 50% of legislators across both chambers belong to governor's party. Middle panel shows sum of coefficients on "Approval" and "Leg. Align × Approval" to represent marginal effect of approval on policy passage for governors aligned with legislature. Column (4) restricts data to only proposals that must move through the legislature to pass, separately from the budget. Standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Included states: AK, AL, CA, CO, FL, GA, HI, IL, IN, KS, MA, MD, MI, MO, MS, NC, ND, NH, NM, NY, OH, OK, TX, UT, VT, WA, WV, WY.

However, columns (3) and (4) show that the *interaction* of legislative alignment and approval is what affects passage: there is a positive effect of gubernatorial approval on policy success, which is driven by governors who are aligned with their legislatures. While the impact of approval by itself is modest and insignificant by itself in these columns (θ_2 is not statistically significant), the marginal effect of approval is significant when looking at the marginal effect for aligned legislatures ($\theta_2 + \theta_4$, shown in the middle panel). In column (3), a one percentage point increase in approval is associated with a 0.56 percentage point increase in likelihood of policy passage. This is consistent with equation (4) in the theoretical section, where the cross-partial of agenda passage with respect to legislative alignment and ability is positive. Columns (3) and (4) additionally add a year fixed effect to capture the effect of the 2001 recession on policy passage.

Finally, column (4) restricts the data to only include policies that must move through the legislature to pass, separately from the budget. In this case, a one standard deviation increase in partisanship leads to a nearly *nine percentage point reduction* in a policy's probability of passage. Moreover, for governors aligned with their legislatures, a one percentage point increase in approval ratings leads to a one percentage point increase in the likelihood a policy from her agenda passes.

5.2 Estimating Model Predictions

We now move to formally test the central behavioral predictions of the model.

- **P1**: When looking at the difference in the partisanship of reelectable and term-limited incumbents, there is a nonmonotonic relationship between approval and partisanship. Unpopular incumbents should pursue relatively partisan agendas, moderately popular incumbents bipartisan agendas, and popular incumbents very partisan agendas.

- **P2**: When increasing legislative alignment, the nonmonotonicity of **P1** should flatten out. However, the minimum level of partisanship of **P1** at moderate levels of win probability should also increase.

- **P3**: At high levels of party entrenchment, the nonmonotonicity of **P1** should flatten out; the nonmonotonicity should be strongest in states that lack party entrenchment.

Let r_{ist} be a binary variable indicating reelection eligibility. This is equal to 1 if a governor is eligible to run another term and 0 otherwise.[26]

Estimation To pinpoint the model's nonmonotonicity, I *bin* approval ratings q_{ist} into deciles by state.[27] Let q_{kist} correspond to the k^{th} decile of approval for a governor in state s. Let X_{st} be a vector of state-level controls; I include the vote share for the governor's party in the previous presidential election (to control for baseline state ideology); the length of the governor's current tenure in office; the state-level unemployment rate (to control for economic shocks); and the total size of the budget. Finally, let ξ_s be a state fixed effect and χ_t a year fixed effect. I estimate the

[26] I count those governors who are eligible for reelection but *choose* not to run again as still being reelection-eligible, as choosing not to run again is an endogenous choice. The central results are robust to counting only governors who actually run for their seats as reelection-eligible.

[27] This accounts for persistent differences in the means of state gubernatorial approval. This also ensures state-level balance in the approval data, allowing the regression to capture departures from average approval *within* each state.

following regression equation.

$$\begin{aligned}\pi_{ist} =& \alpha_0 + \sum_{k\neq 4}^{10} \alpha_k q_{kist} + \gamma_0 \cdot \lambda_{ist} + \sum_{k\neq 4}^{10} \gamma_k q_{kist} \cdot \lambda_{ist} \\ &+ r_{ist} \cdot \big(\beta_0 + \sum_{k\neq 4}^{10} \beta_k q_{kist} + \delta_0 \cdot \lambda_{ist} + \sum_{k\neq 4}^{10} \delta_k q_{kist} \cdot \lambda_{ist}\big) + \omega' X_{st} + \xi_s + \chi_t + \epsilon_{ist}\end{aligned} \tag{9}$$

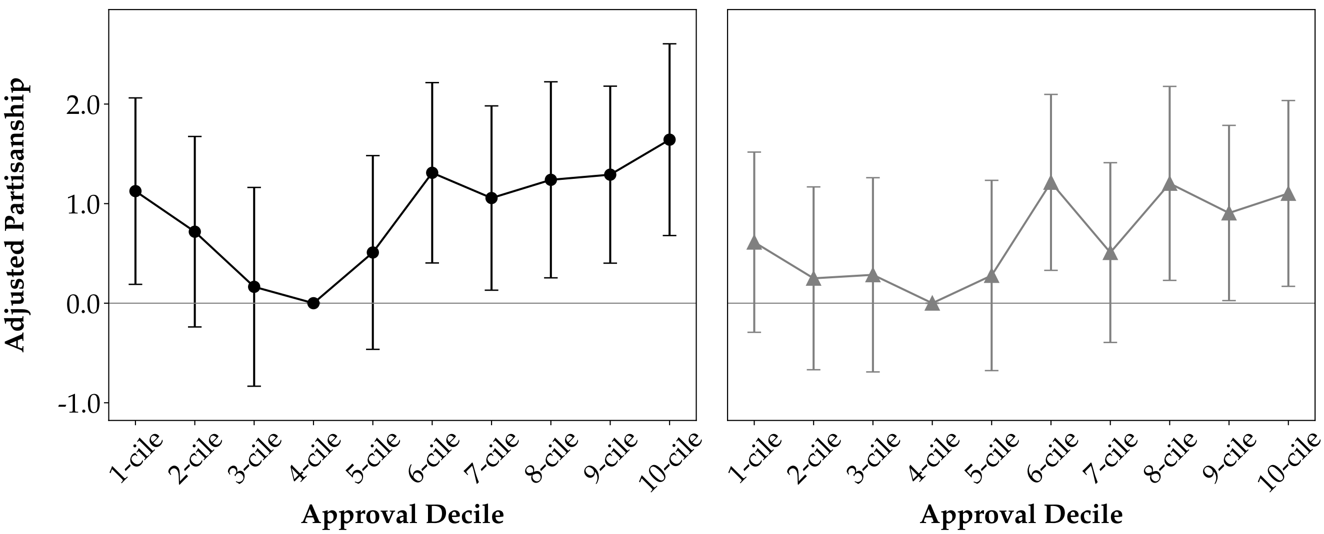
The baseline expression in the first line describes how the level of partisanship varies with approval decile interacted with legislative alignment. The estimands of interest are the β_k and δ_k coefficients, which capture the difference in partisanship for reelection-eligible governors relative to term-limited governors ineligible for reelection. β_k represents partisanship for the k^{th} decile of approval, relative to the fourth decile, for a reelection-eligible governor with an unaligned legislature, adjusting for term-limited governors' behavior.[28] $\beta_k + \delta_k$ represents the level of partisanship for the k^{th} decile of approval, relative to the fourth decile, for a reelection-eligible governor with an *aligned* legislature, adjusting for term-limited behavior.[29]

Results The figure below plots the estimates of the β_k coefficients using black circles in panel (a) on the left. The sums of the $\beta_k + \delta_k$ coefficients are plotted using gray triangles in panel (b) on the right.

[28]I choose the fourth decile as the base category because, as we will see, partisanship experiences a trough in the fourth decile.

[29]The α_k and γ_k coefficients have an analogous interpretation for term-limited governors.

Figure 8: Partisanship of Reelectable Governors by Approval Decile, Adjusting for Term-Limited Behavior, Unaligned vs. Aligned Legislatures, 1990-2020



(a) Unaligned Legislatures

(b) Aligned Legislatures

Dependent variable measures partisanship of U.S. governor State of the State speeches using leave-out estimator of Gentzkow, Shapiro, and Taddy (2019), as calculated in equation (6), normalized by sample mean and standard deviation. Horizontal axis plots approval decile of gubernatorial approval by state, calculated using previous year's fourth quarter approval from Singer (2023) after first year in office, with first quarter approval utilized for first year in office. Black circles plots β_k coefficients from equation (9), measuring level of partisanship for reelectable governors with unaligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Gray triangles plot $\beta_k + \delta_k$, measuring level of partisanship for reelectable governors with aligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Bands around coefficient estimates display 95% confidence intervals. Legislative alignment measured as whether more than half of state legislators match governor's party. Additional controls include vote share for governor's party in most recent presidential election, length of governor's current tenure, state-level unemployment rate, and total state budget expenditures. N: 1108; num. states: 48; within R^2: 0.07. Regression contains year and state fixed effects.

The left panel showcases a nonmonotonic relationship between approval decile and level of partisanship for unaligned legislatures. Being in the first decile of approval means governors' speeches are, on average, 1.1 standard deviations more partisan than in the fourth decile, falling gradually in the second and third deciles. Partisanship relative to the fourth decile begins rising in the fifth decile before remaining at between 1 to 1.5 standard deviations higher in the sixth through tenth deciles. This nonmonotonicity is consistent with prediction **P1** of our model.[30] For reference,

[30] I also look at how approval deciles map onto winning gaps in gubernatorial elections. For governors eligible for reelection in the first approval decile, the average win gap is about six percentage points, i.e. a governor on average wins a race with about 53% of votes. This drops to less than four percentage points for swing states, which I define in the next section, with a large left tail. The average gap rises steadily until about 10 percentage points in the fifth decile and then all the way to 17 percentage points in the top decile. Approval is also positively related to win share; in election years, a one percentage point increase in approval ratings translates to about a 0.4 percentage point increase in winning share.

the overall standard deviation of partisanship in the data is 0.042 percentage points, roughly twice the increase we see in Figure 6 from 2000 to 2020. Figure B.3 in the appendix plots the consequent *levels* of partisanship for reelectable and term-limited governors — i.e. the α_k and $\alpha+\gamma_k$ coefficients and the $\alpha_k + \beta_k$ and $\alpha_k + \beta_k + \gamma_k + \delta_k$ from equation 9. It shows that the nonmonotone pattern in Figure 8(a) emerges as the difference between a shallower (but statistically significant) U pattern for reelectable governors and an upside-down U pattern for term-limited governors.

P2 suggests that when legislative alignment increases, the nonmonotonicity of **P1** flattens out. Although the coefficient estimates seem to display a (muted) U-shape, the lack of any statistically significant relationship between approval and partisanship for aligned legislatures in the right panel is consistent with prediction **P2**. Appendix Figure B.4 shows that this finding is robust to alternative thresholds of legislative alignment. [31]

Table B.2 in the appendix lists the main regression coefficients for equation (9) from which Figure 8 is constructed. It shows that the coefficient on legislative alignment for reelectable governors is positive and statistically significant at the 10% level with a magnitude of 0.367 standard deviations. This is in line with the prediction of **P2** that increases in legislative alignment also cause increases in the baseline level of partisanship.

Appendix Figure B.5 additionally shows that the baseline result is robust to excluding the year reelectable governors are up for reelection; including *only* the year reelectable governors are eligible for reelection; and dropping observations for term-limited governors in their last year in office.[32] The lack of any obvious dynamic effects is in line with the theoretical model's robustness to multiple periods of passage.

Appendix Figure B.6 estimates equation (9) utilizing governor-level fixed effects in addition to year fixed effects. One interpretation is that a governor's fixed effect represents her ideological *bliss point*, meaning the regression equation thus captures departures from this blisspoint. The results are slightly muted but still statistically significant: partisanship for governors with unaligned legislatures drops about 0.7 standard deviations from the first to third decile of approval before rising about one standard deviation as approval reaches the tenth decile.[33]

Finally, Appendix Figure B.7 shows the results are robust to breaking term-limited governors into two groups depending on their presidential aspirations. Panels (a) and (b) use *only* term-limited governors who did not run for president or were not nominated for vice president. The interpretation is that this group of term-limited governors is less likely to exhibit any additional reputational motives in their last term in office, and thus reflects a more accurate counterfactual of what governors would do if they had no electoral concerns. The results in panels (a) and

[31] The magnitude of the results are strongest when defining legislative alignment as equal to 1 if more than 45 or 50% of legislators match the governor's party. The results are weaker using 60%. These findings are consistent with the comparative statics in Corollary 1. The median number of legislators of the governor's party is 55%. The 25^{th} and 75^{th} percentiles are respectively 43% and 65%.

[32] The nonmonotonicity is starkest when dropping term-limited governors in their last year in office since, presumably, these governors have a substantively minimal policy agenda relative to prior years in office.

[33] A limitation of this specification is that governors may systematically face approval ratings grouped in a small set of deciles, meaning a governor fixed effect mechanically dampens the magnitude of the coefficients.

(b) are highly consistent with the benchmark results. Panels (c) and (d) use *only* term-limited governors who *did* run for president or were nominated for vice president. The magnitude of the nonmonotonicity is far starker than in (a) and (b) — a nearly three standard deviation drop in partisanship as governors move from the first to third decile of approval — and is driven by a substantially flatter pattern in the partisanship of governors who lack presidential aspirations.

Disaggregation by Competition The model also predicts, via **P3**, that competitive "swing states" should exhibit more prominent nonmonotonicities than states where a single party always wins the governorship. To test this prediction of the model, I break states into three groups based on the frequency of governor party over the sample period. This allows states to be categorized via a time-invariant definition.

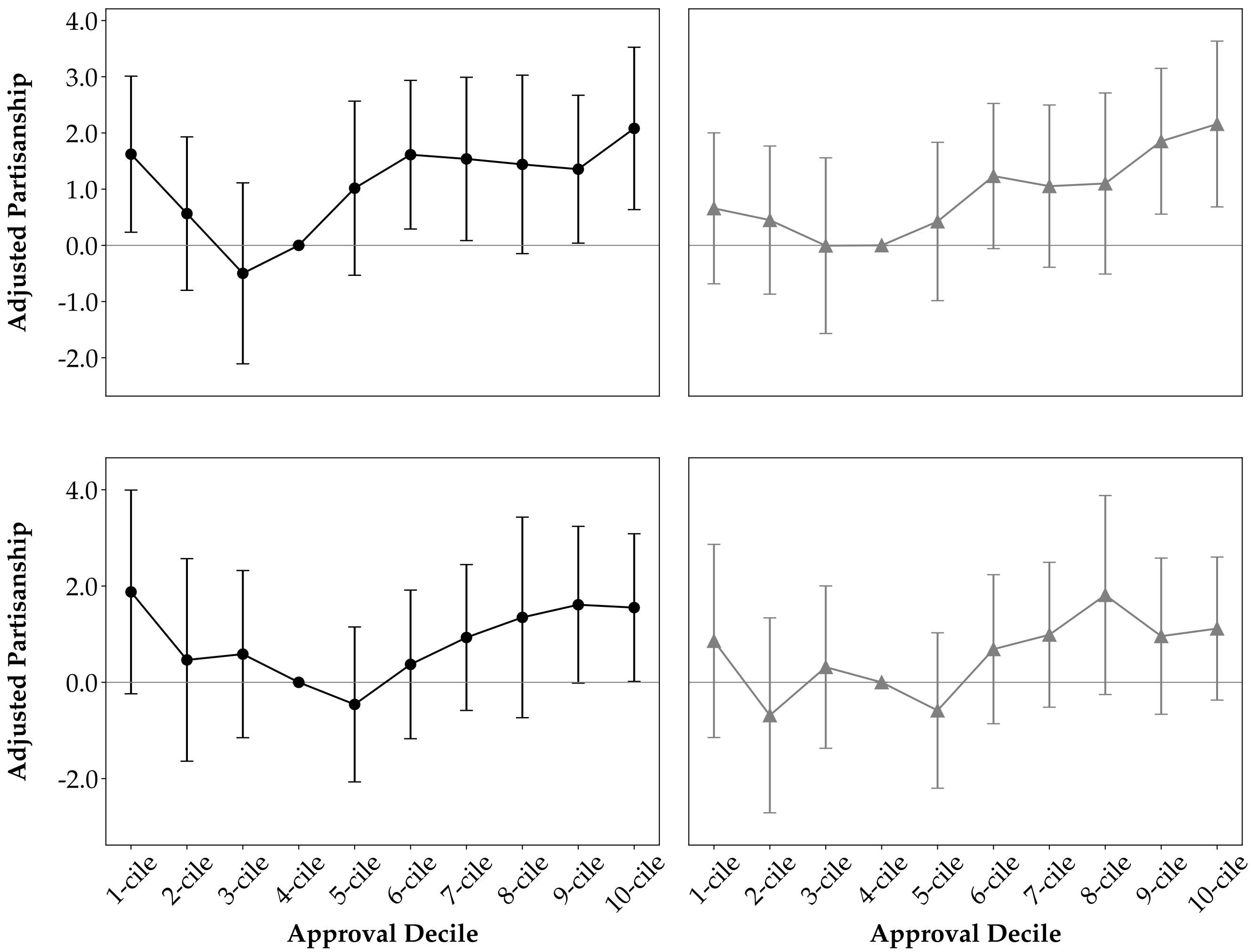
1. Republican States: states with a Republican governor more than 60% of the sample period (1990-2020).
2. Democratic States: states with a Democratic governor more than 60% of the sample period.
3. Swing States: remaining states, where the State governorship experiences fluctuations in party over the sample period.

A list of all states belonging to each category is displayed in Table B.3. The six panels of the figure below showcase estimations of equation (6) for the three categories of state competition: Swing, Republican, and Democratic.

Figure 9: Partisanship of Reelectable Governors by Approval Decile, Adjusting for Term-Limited Behavior, Unaligned vs. Aligned Legislatures, Disaggregated by Competition, 1990-2020

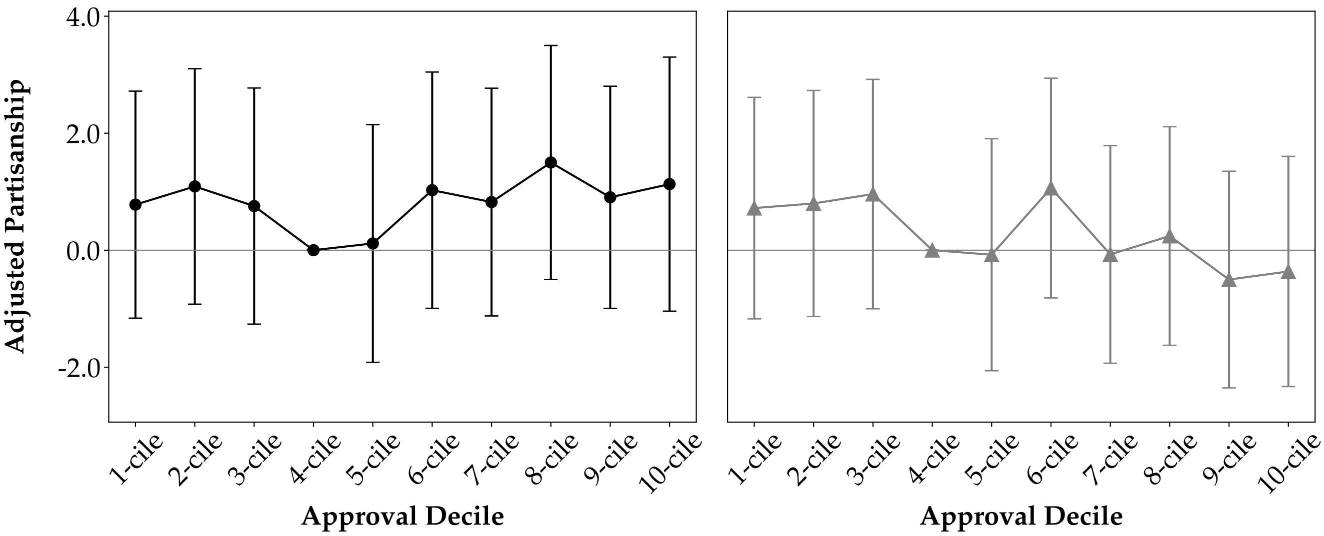
(a) Swing States, Unaligned Legislatures

(b) Swing States, Aligned Legislatures



(c) Rep. States, Unaligned Legislatures

(d) Rep. States, Aligned Legislatures



(e) Dem. States, Unaligned Legislatures

(f) Dem. States, Aligned Legislatures

Dependent variable measures partisanship of U.S. governor State of the State speeches using leave-out estimator of Gentzkow, Shapiro, and Taddy (2019), as calculated in equation (6), normalized by sample mean and standard deviation. Horizontal axis plots approval decile of gubernatorial approval by state, calculated using previous year's fourth quarter approval from Singer (2023) after first year in office, with first quarter approval utilized for first year in office. Black series plots β_k coefficients from equation (9), measuring level of partisanship for reelectable governors with unaligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Gray circles plot $\beta_k + \delta_k$, measuring level of partisanship for reelectable governors with aligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Bands around coefficient estimates display 95% confidence intervals. Legislative alignment measured as whether more than half of state legislators match governor's party. Additional controls include vote share for governor's party in most recent presidential election, length of governor's current tenure, state-level unemployment rate, and total state budget expenditures. Republican states defined as those states with Republican governor more than 60% of sample period. Democratic states defined as those states with Democratic governor more than 60% of sample period. Swing states defined as remaining states. For panels (a) and (b), N: 421; num. states: 18; within R^2: 0.17. For panels (c) and (d), N: 369; num. states: 16; within R^2: 0.22. For panels (e) and (f), N: 318; num. states: 14; within R^2: 0.32.

A nonmonotonicity resembling that in Figure 8 (a) emerges most prominently in panels 9(a) and 9(c), respectively corresponding to swing and Republican states. Partisanship for these states in the first decile of approval is approximately two standard deviations higher than a trough in the third or fifth decile. For swing states, partisanship in the highest decile of approval is about 2.5 standard deviations than the trough, while for Republican states, partisanship is about two standard deviations higher.[34] Consistent with the model, these results are mitigated for governors aligned with their legislatures in panels (b) and (d).

The point estimates for Democratic states in panel (e) exhibit a light nonmonotonicity similar to the swing and Republican states, but this relationship is not statistically significant. To this end, Democratic states provide strong evidence of prediction **P3**: the U-shape of the model should be weaker in states with entrenched parties. While the nonmonotonicity of Republican states in

[34]The asymmetry in partisanship, where the level of partisanship for the lowest approval governors is lower than that of the highest approval governors, is consistent with the general version of the model presented in Theorem 1.

panel (c) is ever so slightly attenuated relative to the swing states in panel (a), one explanation for the more drastic effect for Democrats is that the governors sample tends Republican. States with Democratic governors for more than 60% of the sample comprise 25% of the data, while 36% have Republican governors for more than 60% of the sample. [35]

6 Conclusion

This paper explores how political executives may utilize partisanship to *persuade* the electorate about their political skills and, thereby, how changes in executives' electoral environments may generate variation in partisanship. I develop a theory that interprets an incumbent's choice of policy agenda as an information structure over her ability. Less partisan policies generate left skewness in the distribution of posteriors over ability and more partisan policies generate right skewness. I show that office-motivated incumbents exhibit a nonmonotonic relationship between the partisanship of their policies and their reputation. High and low reputation incumbents pursue partisan agendas, while middling-reputation incumbents pursue bipartisan agendas. These insights are robust to allowing voters to have ideological preferences over partisanship, uncertainty in elections, allowing incumbents to choose opposing parties' platforms, and multiple periods of agenda setting.

I then apply these insights to explain how the partisanship of U.S. governors' policy proposals varies with changes in their electoral environments. I first document that gubernatorial partisanship —measured using governors' annual State of the State speeches — only becomes substantive after the early 2000s, creeps up slowly for the next decade, spikes after 2017, and consistently has a lower level than the comparable series for Congress. I then move to panel data to study partisanship of governors' policy proposals in these speeches, using the model to explore differences in the partisanship of reelectable and term-limited governors. Using passage data from Kousser and Phillips (2012), I show that more partisan agendas are less likely to pass, and that governor approval is positively associated with policy passage for governors aligned with their legislatures. I show that a nonmonotonic variation of gubernatorial partisanship with approval decile in unaligned legislatures matches that of the model, especially in swing states. In line with the model, increasing legislative alignment flattens this nonmonotonicity; and that highly Democratic states are also less likely to showcase this nonmonotonicity. The panel analysis coupled with the theoretical analysis permits a view that changes in partisanship as not solely driven by latent trends in political landscapes but also shaped by elected officials' reelection incentives.

The paper broadly advances three agendas. First, the model links the documentary literature on partisanship to the theoretical literature on political accountability. It predicts a nonmononotic relationship between popularity and partisanship that matches patterns in the data, unlike much of the literature's "gambling for resurrection" models. Second, the paper provides a novel series

[35] While a more strict criterion for Republican states is appealing — such as, e.g., defining them as states with Republican governors for more than 70% of the sample — a definition like this suffers from missing values for some term-limited categories.

measuring partisanship of political executives by utilizing governors' State of the State speeches. It also provides a novel series measuring partisanship of political speech at the state level. Finally, the paper links the theoretical framework to aggregate partisanship with specific attention to explaining *variations* in partisanship, contrasting with a large literature which has largely documented *trends* in partisanship. The model does not interpret heterogeneity in partisanship as state-level noise, but as part of a structured response to electoral incentives.

In the dataset at hand, partisan attitudes manifest themselves in efforts to create affordable housing, combat climate change, restructure law enforcement, fund charter schools, extend healthcare coverage, raise minimum wages, create hospitable business environments, or cut taxes. These policies have large effects on the shape of economic inequality, social stratification, and human capital acquisition within the States. Providing insight on when, how, and why politicians may pursue more or less partisan agendas permits richer insights into the timing, nature, and potential effects of these sorts of policies on the landscape of partisanship in American politics.

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Appendix A Proofs and Theoretical Extensions

A.1 Main Proofs

Proposition 1. *There exist thresholds $0 < \underline{q} < q_L^2 < \overline{q} < 1$ such that:*

- *for $q_R^1 \in [0, \underline{q})$, $\pi^*(q_R^1)$ is strictly decreasing, with $\pi^*(0) = 1$ and $\pi^*(\underline{q}) = 0$;*
- *for $q_R^1 \in [\underline{q}, \overline{q})$, $\pi^*(q_R^1) = 0$;*
- *for $q_R^1 \in [\overline{q}, 1]$, $1 \in \pi^*(q_R^1)$ with equality at $\overline{q}$.*

Proof. Note in general the following formulae for posteriors as a function of π, fixing q_R^1:

$$\overline{q}_R^2(\pi) = \frac{(\lambda + (1-\lambda)(1-\pi))q_R^1}{\lambda q_R^1 + (1-\lambda)(1-\pi)}$$

$$\underline{q}_R^2(\pi) = \frac{(1-\lambda)\pi q_R^1}{\lambda(1-q_R^1) + (1-\lambda)\pi}.$$

Note that $\overline{q}_R^2$ is decreasing in π while $\underline{q}_R^2$ is increasing in π.

Next, fix $q_R^1 \in (0,1)$. Consider the following sets $\overline{P}$ and $\underline{P}$:

$$\underline{P}(q_R^1) := \{\pi : \overline{q}_R^2(\pi) \geq q_L^2, \underline{q}_R^2(\pi) < q_L^2\}$$

$$\overline{P}(q_R^1) := \{\pi : \underline{q}_R^2(\pi) \geq q_L^2\}.$$

Note that $\underline{P}$ is always nonempty ($\ni \pi = 1$) but $\overline{P}$ is empty for q_R^1 low. I claim that $\pi^*(q_R^1) = \min\{\underline{P}(q_R^1)\}$ when $\overline{P} = \varnothing$ and $\overline{P}$ otherwise.

First, any $\pi \notin \overline{P}, \underline{P}$ always leads to an expected win probability of 0; these are dominated by any $\pi \in \underline{P}$. Next, for any $\pi \in \underline{P}$, the expected win probability is $\lambda q_R^1 + (1-\lambda)(1-\pi)$, which is the likelihood of seeing $\overline{q}_R^2$. This probability is maximized when π is minimized, i.e. at $\min\{\underline{P}(q_R^1)\}$. If $\overline{P}$ is empty, then $\pi^*(q_R^1) = \min\{\underline{P}(q_R^1)\}$. Finally, if $\overline{P}$ is nonempty, any $\pi \in \overline{P}$ leads to a win with probability 1, meaning $\pi^*(q_R^1) = \overline{P}(q_R^1)$.

Next, we show $\min\{\underline{P}(q_R^1)\}$ is strictly decreasing in π until a point $\underline{q}$, whereafter it is equal to 0. In general, $\min\{\underline{P}(q_R^1)\}$ is the solution to $\frac{(\lambda+(1-\lambda)(1-\pi))q_R^1}{\lambda q_R^1 + (1-\lambda)(1-\pi)} = q_L^2$. As q_R^1 rises, the left-hand-side increases for each π, meaning the π solving this equation decreases. The solution to this equation exists up until some $\underline{q} < q_L^2$, defined by $\frac{(\lambda+(1-\lambda))\underline{q}}{\lambda\underline{q}+(1-\lambda)} = q_L^2$, where $\pi = 0$ and can no longer decrease. For $q_R^{1'} > q_R^1$, if $\pi \in \underline{P}(q_R^1)$, then $\pi \in \underline{P}(q_R^{1'})$, meaning $\min\{\underline{P}(q_R^1)\} = 0$ for $q_R^1 \geq \underline{q}$. Finally, as $q_R^1 \to 0$, $\pi \to 1$.

Finally, we show that there exists $\overline{q} > q_L^2$ such that $\overline{P}(q_R^1) \neq \varnothing$ if and only if $q_R^1 \geq \overline{q}$. Note that, fixing q_R^1, the expression for $\underline{q}_R^2$ is maximized when $\pi = 1$. This means that $\overline{P}$ is nonempty if and only if $\frac{(1-\lambda)q_R^1}{1-\lambda q_R^1} \geq q_L^2$, which occurs if and only if $q_R^1 \geq \overline{q} > q_L^2$ defined as the implicit solution to $\frac{(1-\lambda)\overline{q}}{1-\lambda\overline{q}} = q_L^2$.

The comparative statics with respect to q_L^2 and λ emerge directly from the equations defining $\overline{q}$ and $\underline{q}$. $\square$

Proposition 2. *Suppose the voter can choose legislative alignment λ, and his second period utility is given by $U_R^2(a_R, \pi_R)$. Then, U_R^2 is increasing in a_R for $\pi_R > b$ if and only if*

$$\delta \leq \frac{\pi_R}{\pi_R b - b^2} \equiv \overline{\delta}.$$

Moreover, $\overline{\delta} \geq \frac{4}{\pi_R}$.

Proof. Let λ^* denote the voter's choice of λ. Since $\pi_R \geq b$, we always have $\lambda^* \pi_R \leq b$. Recall that $p(\lambda, a_R, \pi_R) = \lambda a_R + (1 - \lambda)(1 - \pi_R)$. This means U_R^2 can be written as

$$a_R - \delta\big(p(\lambda, a_R, \pi_R)(\pi_R - b) + (1 - p(\lambda, a_R, \pi_R))(b - \lambda\pi_R)\big).$$

By the Envelope Theorem, the derivative of U_R^2 with respect to a_R is given by

$$1 - \delta\big[(p_{a_R}(\lambda, a_R, \pi_R)\big((\pi_R - b) - (b - \lambda^*\pi_R)\big)\big] = 1 - \delta\lambda^*\big((1 + \lambda^*)\pi_R - 2b\big),$$

which is $\geq 0 \iff \delta \leq \frac{1}{\lambda^*\big((1+\lambda^*)\pi_R - 2b\big)}$. Next, we characterize λ^*, which minimizes

$$G(\lambda, a_R, \pi_R) := p(\lambda, a_R, \pi_R)(\pi_R - b) + (1 - p(\lambda, a_R, \pi_R))(b - \lambda\pi_R)\big),$$

which is quadratic in λ. The first derivative with respect to λ is given by

$$p_\lambda(\lambda, a_R, \pi_R)((1 + \lambda)\pi_R - 2b) - \pi_R(1 - p(\lambda, a_R, \pi_R)).$$

Since $p_\lambda = a_R - (1 - \pi)$, we have $p_{\lambda\lambda} = 0$. Thus, the second derivative is:

$$p_{\lambda\lambda}(\lambda, a_R, \pi_R)((1 + \lambda)\pi_R - 2b) + 2\pi_R p_\lambda(\lambda, a_R, \pi_R) = 2\pi_R(a_R - (1 - \pi_R)).$$

The function is hence convex if $a_R \geq 1 - \pi_R$ and concave if $a_R \leq 1 - \pi_R$. We either have $a_R = 1$ (the former case) or $a_R = 0$ (the latter).

We start with $a_R = 1$. With some abuse of notation, we write $p(\lambda, a_R, \pi_R) = \lambda p_\lambda + 1 - \pi_R$ and simplify the first order condition as

$$\begin{aligned} p_\lambda((1+\lambda)\pi_R - 2b) - \pi_R(1-(\lambda p_\lambda + 1 - \pi_R)) = p_\lambda((1+\lambda)\pi_R - 2b + \lambda\pi_R) - \pi_R^2 &= 0 \\ p_\lambda((1+2\lambda)\pi_R - 2b) &= \pi_R^2 \\ (1+2\lambda)\pi_R &= 2b + \frac{\pi_R^2}{a_R - (1-\pi_R)} \\ \lambda^* &= \frac{b}{\pi_R} + \frac{\pi_R}{2(a_R - (1-\pi_R))} - \frac{1}{2} \\ &= \frac{b}{\pi_R}. \end{aligned}$$

Now, consider the case where $a_R = 0$, which leads to a corner solution $\lambda = 0$ or $\lambda = \frac{b}{\pi_R}$. We have

$$G(0,0,\pi_R)(1-\pi_R)(\pi_R - b) + \pi_R b \geq (1 - \frac{b}{\pi_R})(1-\pi_R)(\pi_R - b) = G(\frac{b}{\pi_R}, 0, \pi_R).$$

Hence, $\lambda^* = \frac{\pi}{b_R}$ in all cases, and U_R^2 is increasing in a_R if and only if

$$\delta \leq \frac{\pi_R}{\pi_R b - b^2} = \frac{1}{\frac{b}{\pi_R}(\pi_R - b)} = \frac{\pi_R}{\pi_R b - b^2}.$$

Fixing b, this expression is decreasing in π_R, and achieves a minimum with respect to b at $b = \frac{\pi_R}{2}$. ☐

Proposition 3. *As $q_L^2 \to 1$, $\underline{q} \to \overline{q}$. As $q_L^2 \to 0$, $\overline{q} \to \underline{q}$. There exists q_L^{2*} that maximizes the size of the interval $[\underline{q}, \overline{q})$.*

Proof. The closed forms for $\underline{q}, \overline{q}$ are respectively, based on a previous proposition:

$$\frac{(1-\lambda)q_L^2}{1-\lambda q_L^2}, \frac{q_L^2}{1-\lambda(1-q_L^2)}.$$

Both terms are increasing in q_L^2. The derivative of the difference of these two terms with respect to q_L^2 is given by:

$$\frac{1-\lambda}{(1-\lambda(1-q_L^2))^2} - \frac{1-\lambda}{(1-\lambda q_L^2)^2}$$

which is positive for $q_L^2 \leq 1/2$, negative when $q_L^2 \geq 1/2$, and $= 0$ at $q_L^{2*} = 1/2$. ☐

The following two lemmata are used to prove Theorem 1.

Lemma 1. *Suppose w is a differentiable, strictly concave, and strictly increasing function defined over q_R^2. Then, there exists $p \in [0,1]$ such that the expected value of w over the lottery of posteriors $\overline{q}_R^2, \underline{q}R^2$ is maximized when $(p_1, p_0) = (p, p)$.*

Proof. We make a slight change of notation, $q_R^1 = q$. Our problem describing the maximum of w over the lotteries over q_R^2 is given by:

$$\max_{p_1 \geq p_0} \int_{q_R^2} w(q_R^2) dF(q_R^2 | p_1, p_0, q).$$

Let $g(p_1)$ be the probability of a success conditional on $a_R = 1$; and $h(p_0)$ the probability of success conditional on $a_R = 0$. Note that the distribution of posteriors $F(q_R^2 | p_1, p_0, q_R^2)$ is a two-point mean preserving spread of the prior q.

Suppose by contradiction that $p_1 > p_0$. This means that there would exist p_1, p_0 such that

$$\begin{aligned}\overline{q}_R^2(p_1, p_0) &= \frac{g(p_1)q}{g(p_1)q + h(p_0)(1-q)} \\ \underline{q}_R^2(p_1, p_0) &= \frac{q - qg(p_1)}{1 - qg(p_1) - (1-q)h(p_0)}.\end{aligned}$$

The partial derivatives of $\overline{q}_R^2$ are:

$$\begin{array}{ll} p_1 & q(1-q)g'(p_1)\dfrac{h(p_0)}{[g(p_1)q + h(p_0)(1-q)]^2} > 0 \\ p_0 & -q(1-q)h'(p_0)\dfrac{g(p_1)}{[g(p_1)q + h(p_0)(1-q)]^2} < 0. \end{array}$$

The partial derivatives of $\underline{q}_R^2$ are:

$$\begin{array}{ll} p_1 & -q(1-q)g'(p_1)\dfrac{(1-h(p_0))}{[1 - qg(p_1) - (1-q)h(p_0)]^2} < 0 \\ p_0 & q(1-q)h'(p_0)\dfrac{(1-g(p_0))}{[1 - qg(p_1) - (1-q)h(p_0)]^2} > 0 \end{array}$$

Suppose we marginally decrease p_1 by Δ_1 and increase p_0 by Δ_0 so that $\overline{q}_R^2$ remains the same, i.e.

$$\begin{aligned} q(1-q)g'(p_1)\frac{h(p_0)}{[g(p_1)q + h(p_0)(1-q)]^2}\Delta_1 &= q(1-q)h'(p_0)\frac{g(p_1)}{[g(p_1)q + h(p_0)(1-q)]^2}\Delta_0 \\ &\implies g'(p_1)h(p_0)\Delta_1 - g(p_1)h'(p_0)\Delta_0 = 0 \end{aligned}$$

The sign of the change in $\underline{q}_R^2$ is then the sign of:

$$\begin{aligned} &-g'(p_1)(1-h(p_0))\Delta_1 + h'(p_0)(1-g(p_1)\Delta_0 \\ &= g'(p_1)h(p_0))\Delta_1 - g(p_1)h'(p_0)\Delta_0 - g'(p_1)\Delta_1 + h'(p_0)\Delta_0 \\ &= -g'(p_1)\Delta_1 + h'(p_0)\Delta_0 \end{aligned}$$

$g(p_1)$ is given by $\lambda + (1-\lambda)p_1$, so its derivative is $(1-\lambda)$. $h(p_0)$ is given by $(1-\lambda)p_0$, so its derivative is also $(1-\lambda)$. Hence, the sign of the change in $\underline{q}_R^2$ is the sign of $\Delta_0 - \Delta_1$.

I claim that $\Delta_0 > \Delta_1$. The expressions for these are

$$\begin{aligned}\Delta_1 &= q(1-q)g'(p_1)h(p_0)\\ \Delta_0 &= q(1-q)g(p_1)h'(p_0).\end{aligned}$$

The latter is larger than the former if and only if

$$g(p_1)h'(p_0) > g'(p_1)h(p_0).$$

Since $h'(p_0) = g'(p_1) = 1-\lambda$, this is true if and only if $g(p_1) > h(p_0)$, which is always true by the constraint that $p_1 \geq p_0$ (i.e. the conditional likelihood of a success is higher for $a_R = 1$ than $a_R = 0$). Hence, decreasing p_1 by Δ_1 and raising p_0 by Δ_0 keeps $\overline{q}_R^2$ fixed while raising $\underline{q}_R^2$, generating a mean preserving contraction of the original lottery over posteriors. Because w is strictly increasing and concave, this new lottery over posteriors is preferred to the original generated by (p_1, p_0), a contradiction. Hence, we always have that w is maximized when $p_1 = p_0 = p$. □

Lemma 2. *Suppose w is a differentiable, strictly concave, and strictly increasing function defined over q_R^2. Then, the lottery generated by $(p,p) = (0,0)$ over q_R^2 dominates all other (p_1, p_0).*

Proof. Note first that $\overline{q}_R^2$ is strictly decreasing as a function of p, meaning $\overline{q}_R^2$ is maximized at $p = 0$. Note that any lottery over posteriors can be represented with a line segment connecting $(\underline{q}_R^2, w(\underline{q}_R^2))$ and $(\overline{q}_R^2, w(\overline{q}_R^2))$, with the expected value of the lottery given by the point on the segment corresponding to the prior q.

Note that, fixing $\overline{q}_R^2$, flatter line segments correspond to higher expected values. It then suffices to show that the line segment connecting $(\underline{q}_R^2, w(\underline{q}_R^2))$ and $(\overline{q}_R^2, w(\overline{q}_R^2))$ is shallowest at $p = 0$ — where $w(\overline{q}_R^2)$ is additionally maximal. I.e., its slope, given by

$$\frac{w(\overline{q}_R^2(p)) - w(\underline{q}_R^2(p))}{\overline{q_R^2} - \underline{q}_R^2},$$

achieves a minimum at $p = 0$. The numerator of the derivative of this expression is given by:

$$[\overline{q_R^2} - \underline{q}_R^2][w'(\overline{q}_R^2)\overline{q_R^2}' - w'(\underline{q}_R^2)\underline{q_R^2}'] - [w(\overline{q}_R^2) - w(\overline{q}_R^2)][\overline{q}_R^{2\prime} - \underline{q}_R^{2\prime}]$$

Note that $h'(p) = g'(p) = (1-\lambda)$, and that $g(p) - h(p) = \lambda$, so that:

$$\begin{aligned}\overline{q}_R^{2\prime} - \underline{q}_R^{2\prime} &= \frac{q(1-q)[g'(p)h(p) - h'(p)g(p)]}{[qg(p) + (1-q)h(p)]^2} - \frac{q(1-q)[g'(p)(1-h(p)) - h'(p)(1-g(p))}{[1 - qg(p) - (1-q)h(p)]^2}\\ &= \frac{q(1-q)\lambda(1-\lambda)}{[1 - qg(p) - (1-q)h(p)]^2} - \frac{q(1-q)\lambda(1-\lambda)}{[qg(p) + (1-q)h(p)]^2}\end{aligned}$$

Hence the derivative above is ≥ 0 if and only if

$$
\begin{aligned}
&[\overline{q}^2_R - \underline{q}^2_R][-w'(\overline{q}^2_R)\frac{q(1-q)\lambda(1-\lambda)}{[qg(p)+(1-q)h(p)]^2} + w'(\underline{q}^2_R)\frac{q(1-q)\lambda(1-\lambda)}{[1-qg(p)-(1-q)h(p)]^2}] \\
&\quad -[w(\overline{q}^2_R) - w(\overline{q}^2_R)]\left[\frac{q(1-q)\lambda(1-\lambda)}{[1-qg(p)-(1-q)h(p)]^2} - \frac{q(1-q)\lambda(1-\lambda)}{[qg(p)+(1-q)h(p)]^2}\right] \geq 0 \\
&\frac{w'(\underline{q}^2_R)}{[1-qg(p)-(1-q)h(p)]^2} - \frac{w'(\overline{q}^2_R)}{[qg(p)+(1-q)h(p)]^2} \\
&\geq \frac{\frac{w(\overline{q}^2_R)-w(\overline{q}^2_R)}{\overline{q}^2_R-\underline{q}^2_R}}{[1-qg(p)-(1-q)h(p)]^2} - \frac{\frac{w(\overline{q}^2_R)-w(\overline{q}^2_R)}{\overline{q}^2_R-\underline{q}^2_R}}{[qg(p)+(1-q)h(p)]^2}
\end{aligned}
$$

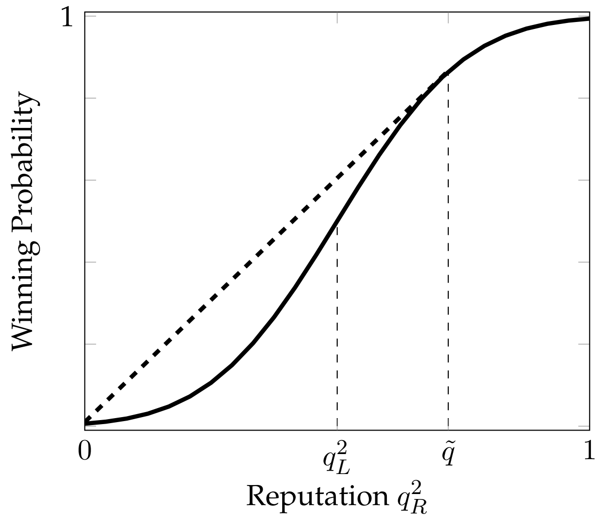
Note in particular that by concavity, $w'(\underline{q}^2_R) > \frac{w(\overline{q}^2_R)-w(\overline{q}^2_R)}{\overline{q}^2_R-\underline{q}^2_R} > w'(\underline{q}^2_R)$. This means that the first term on the left is always strictly larger than the first time on the right; and that the magnitude of the second term on the left is smaller than the magnitude of the second term on the right, so that this expression is always true, i.e. the derivative is positive for all p. Hence, the slope of the segment achieves a minimum at $p = 0$, showing the result. $\square$

Theorem 1. *There exist thresholds $\underline{q} < \overline{q}$ such that:*

- *for $q^1_R \leq \underline{q}$, $\pi^*(q^1_R)$ is strictly decreasing. $p^*(0) = (1,0)$ and $\pi^*(0) = 1/2$, while $\pi^*(\underline{q}) = 0$ and $\pi^*(\underline{q}) = 0$.*
- *For $q^1_R \in [\underline{q}, \overline{q})$, $p^* = (1,1)$ and $\pi^* = 0$;*
- *for $q^1_R \geq \overline{q}$, $p^* = (0,0)$ and $\pi^* = 1$.*

Proof. The proof of this result makes use of the previous two lemmata. First, note that the concavification of $N(q^2_R - q^2_L)$ is characterized by a point $\tilde{q} > q^2_L$. Specifically, the concavification is given by a line segment connection the points $(0, N(-q^2_L))$ and $(\tilde{q}, N(\tilde{q} - q^2_L))$, where $\tilde{q}$ solves $N'(\tilde{q} - q^2_L)\tilde{q} = N(\tilde{q} - q^2_L) - N(-q^2_L)$, followed by the curve N itself for $q \geq \tilde{q}$. Note also that N is strictly concave for $q^1_R > q^2_L$ and strictly convex for $q^1_R < q^2_L$. These are shown in the figure below, where the solid line is $N(q^2_R - q^2_L)$ and the dashed line the concavification.

Figure A.1: Concavification of $N(q_R^2 - q_L^2)$



Next, notice that there exists $0 < \underline{q} < \tilde{q}$ such that for all $q_R^1 \leq \underline{q}$, we can choose p_1 and p_0 to achieve the concavification. Since $\underline{q}_R^2$ must = 0 for any point $\leq \tilde{q}$ to achieve the concavification, we must have $p_1 = 1$ and vary p_0. p_0 must then solve, for each q_R^1,

$$\frac{q_R^1}{q_R^1 + (1-\lambda)(1-q_R^1)p_0} = \tilde{q}.$$

Notice that this expression is minimized when $p_0 = 1$, when it is equal to $\frac{q_R^1}{q_R^1+(1-\lambda)(1-q_R^1)}$. Let $\underline{q}$ solve $\frac{\underline{q}}{\underline{q}+(1-\lambda)(1-\underline{q})} = \tilde{q}$. Note that for all $q_R^1 > \underline{q}$, there does not exist $p_0 \in [0,1]$ such that the posteriors $(\overline{q}_R^2, \underline{q}_R^2) = (\tilde{q}, 0)$ are achievable. It is easy to see that for all $q \leq \underline{q}$, there exists $p_0(q_R^1)$ that achieves these posteriors, with p_0 given directly by

$$p_0 = \frac{q_R^1(1-\tilde{q})}{(1-\lambda)(1-q_R^1)\tilde{q}},$$

which is strictly increasing from 0 at $q_R^1 = 0$ to 1 at $\underline{q}$. This argument uses a similar approach to Proposition 1.[36]

Next, we show that for all $q_R^1 \geq \underline{q}$, $(p_1, p_0) = (1,1)$ dominates any (p_1, p_0) such that $\underline{q}_R^2 < q_L^2$. Suppose by contradiction that there exists $(p_1, p_0) \neq (1,1)$ such that $\underline{q}_R^2 < q_L^2$, which dominates $(1,1)$. Because $\overline{q}_R^2$ is minimized at $(p_1, p_0) = (1,1)$, we necessarily have that $\overline{q}_R^2 > q_L^2$, so that $\overline{q}_R^2$ is on the concave portion of N. Suppose first that $p_1 = 1$ so that $\underline{q}_R^2 = 0$. The expected value of this lottery is the line segment from $(0, N(-q_L^2))$ to $(\overline{q}_R^2, N(\overline{q}_R^2 - q_L^2))$. Since $p_0 < 1$, we can generate a strict improvement by increasing p_0, which slightly lowers $\overline{q}_R^2$. However, because $\overline{q}_R^2$ is on the concave portion of N, the line segment from $(0, N(-q_L^2))$ to $(\overline{q}_R^2, N(\overline{q}_R^2 - q_L^2))$ becomes steeper

[36]In particular, $\frac{\underline{q}}{\underline{q}+(1-\lambda)(1-\underline{q})} = \tilde{q}$ gives the comparative static of $\underline{q}$ with respect to λ and q_L^2. Increasing λ decreases $\underline{q}$. Increasing q_L^2 increases $\tilde{q}$ and hence increases $\underline{q}$.

while originating from the same point $(0, N(-q_L^2))$. Because the expected value from (p_1, p_0) lies on the point on the segment corresponding to q_1^2, this steepening generates an improvement on the original (p_1, p_0), a contradiction.

Hence, suppose $p_1 < 1$ so that $\underline{q}_R^2 > 0$. If $p_0 > 0$, we can reverse the argument in Lemma 1 and slightly raise p_1 and lower p_0 so that $\overline{q}_R^2$ remains the same but $\underline{q}_R^2$ decreases. The line segment connecting $(\underline{q}_R^2, N(\underline{q}_R^2 - q_L^2))$ to $(\overline{q}_R^2, N(\overline{q}_R^2 - q_L^2))$ becomes shallower, while its upper point $(\overline{q}_R^2, N(\overline{q}_R^2 - q_L^2))$ remains the same, meaning the line rotates upwards, again generating an improvement. If $p_0 = 0$, increasing p_0 causes both $\underline{q}_R^2$ and $\overline{q}_R^2$ to decrease. However, because $\overline{q}_R^2$ is on the concave portion of N and $\underline{q}_R^2$ on the convex portion, this generates a left/upward shift in the line segment connecting $(\underline{q}_R^2, N(\underline{q}_R^2 - q_L^2))$ to $(\overline{q}_R^2, N(\overline{q}_R^2 - q_L^2))$, again generating an improvement.

Next, we show that there exists $\overline{q}$ such that for all $q_R^1 \geq \overline{q}$, $(p_1, p_0) = (0,0)$ dominates $(1,1)$. $(0,0)$ generates posteriors $\overline{q}_R^2 = 1$ and $\underline{q}_R^2 = \frac{(1-\lambda)q_R^1}{1-\lambda q_R^1}$. Note that because $1 > \frac{q_R^1}{q_R^1+(1-\lambda)(1-q_R^1)}$, a sufficient condition for this to hold is that the line segment connecting and $(\frac{(1-\lambda)q_R^1}{1-\lambda q_R^1}, N(\frac{(1-\lambda)q_R^1}{1-\lambda q_R^1} - q_L^2)$ and $(1, N(1-q_L^2))$ is shallower than that connecting $(0, N(-q_L^2))$ to $(\frac{q_R^1}{q_R^1+(1-\lambda)(1-q_R^1)}, N(\frac{q_R^1}{q_R^1+(1-\lambda)(1-q_R^1)} - q_L^2))$. As $q_R^1 \to 1$, the slope of the former line segment approaches 0; while the slope of the latter approaches $N(1-q_L^2) - N(-q_L^2)$. Because the change in these slopes is monotone as long as q_R^1 is sufficiently high (i.e. as long as $\frac{(1-\lambda)q_R^1}{1-\lambda q_R^1} \geq q_L^2)$, by continuity, there exists $\overline{q}$ such that for $q_R^1 \geq \overline{q}$, $(0,0)$ dominates $(1,1)$.

Finally, using Lemma 2, we know that $(0,0)$ dominates any other (p_1, p_0) such that $\overline{q}_R^2, \underline{q}_R^2 \geq q_L^2$. Because $(1,1)$ dominates any (p_1, p_0) with $\overline{q}_R^2 \geq q_L^2 \geq \underline{q}_R^2$, $(0,0)$ also dominates all these points, showing the result.

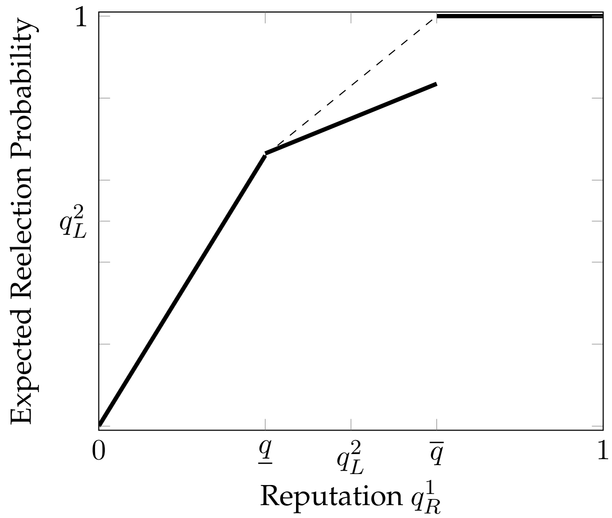
$\square$

Proposition 4. *Given $\underline{q}, \overline{q}$, there exist thresholds $\underline{\underline{q}} < \underline{q} < \overline{q} < \overline{\overline{q}}$ such that*

- *For $q_R^0 \leq \underline{\underline{q}}$, partisanship π^* is decreasing from $1/2$ at q_R^0 to 0 at $\underline{\underline{q}}$.*
- *There exists $q_\dagger \in [\underline{\underline{q}}, \overline{q} < \overline{\overline{q}}]$ such that for all $q_R^0 \geq q_\dagger$, $\pi^* < 1$.*
- *For $q_R^0 \geq \overline{\overline{q}}$, $\pi^* = 1$.*

Proof. We graph the value function of R, $V_R(q_R^1)$, as a function of q_R^1 in the figure below.

Figure A.2: Value Function of R As Function of q_R^1



The concavification of the value function is achievable for low q_R^1 as a lottery between the beliefs 0 and $\underline{q}$, achieved via $(p_1, p_0) = (1, p_0^*)$ for p_0^* solving $\frac{q_R^0}{q_R^0 + (1-q_R^0)(1-\lambda)p_0^*} = \underline{q}$. p_0^* increases in q_R^0 until some $\underline{\underline{q}}$, when it is equal to 1. By a similar argument from before, partisanship $\pi^* = 1 - \frac{1+p_0^*}{2}$ is then decreasing from 1/2 at 0 to 0 at $\underline{\underline{q}}$. An identical argument to proposition 1 shows that above some threshold $\overline{\overline{q}}$, the optimal agenda is given by $(p_1, p_0) = (0, 0)$.

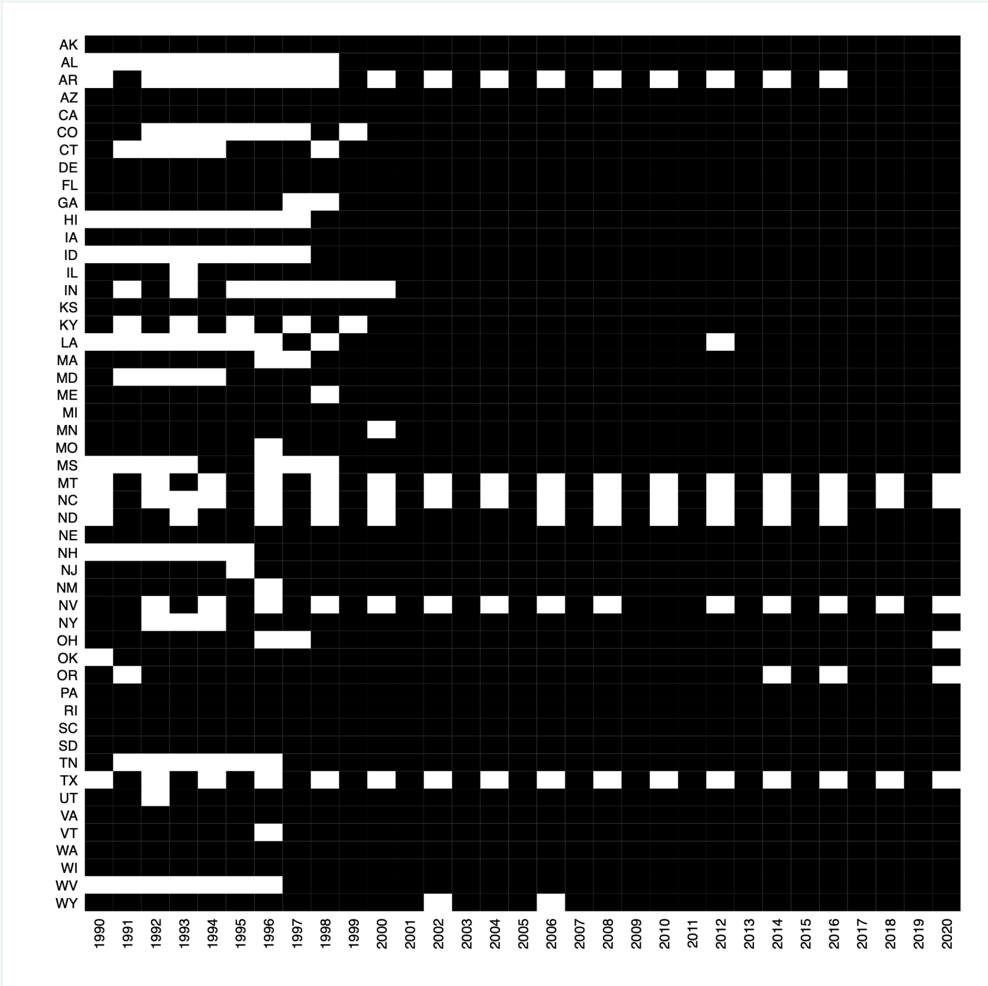
Let $q_\dagger$ solve $\underline{q} = \frac{(1-\lambda)q_R^0}{1-\lambda q_R^0}$. Note that for each (p_1, p_0), we have a line segment connecting $(\underline{q}_R^1, V_R(\underline{q}_R^1))$ to $(\overline{q}_R^1, V_R(\overline{q}_R^1))$. Moreover, for $q_R^0 \in [q_\dagger, \overline{\overline{q}}]$, the expected value of $(p_1, p_0) = (0, 0)$ is the segment connecting the value function at $\underline{q}$ to the value function at 1 (i.e. an extension of the value function on $[\underline{q}, \overline{q}]$).

I claim that for all $q_R^0 \in (q_\dagger, \overline{\overline{q}})$, $\pi^* < 1$, i.e. $(p_1, p_0) \neq (0, 0)$. To see this, note that by slightly increasing both p_1 and p_0, $(\underline{q}_R^1, V_R(\underline{q}_R^1))$ decreases linearly. However, $(\overline{q}_R^1, V_R(\overline{q}_R^1))$ slides to the left without decreasing, meaning the line segment steepens and, at the prior q_R^0, generates an improvement on $(p_1, p_0) = (0, 0)$. The expressions for points in between are generally dependent on λ and q_L^2.

$\square$

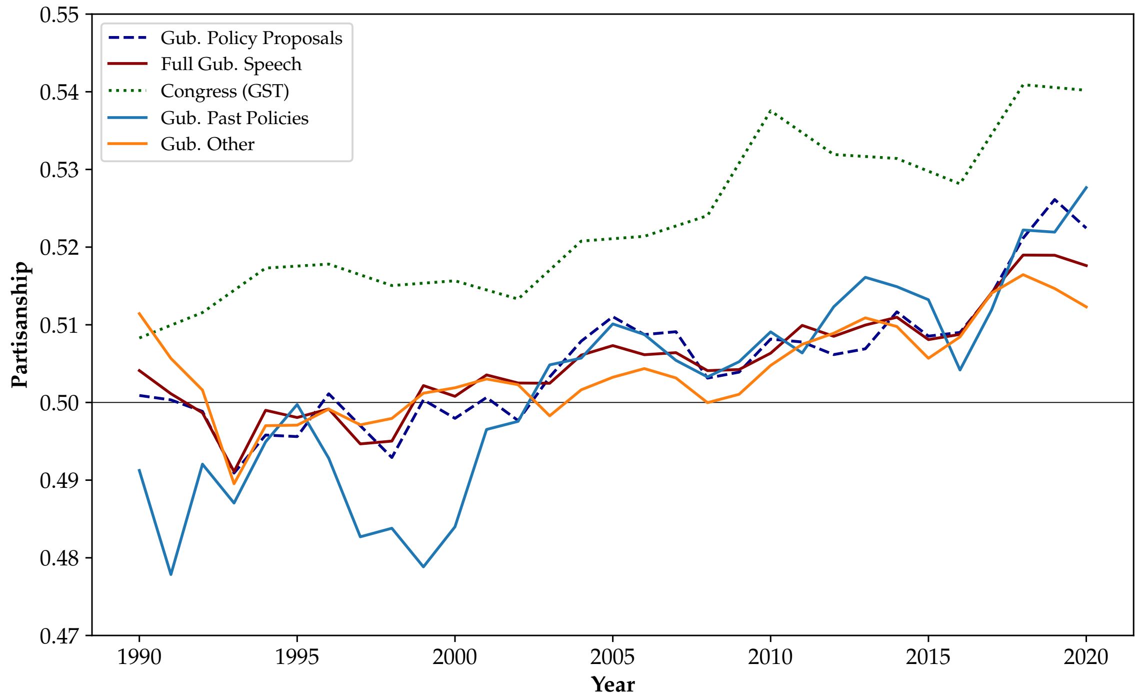
Appendix B Auxiliary Tables and Figures

Figure B.1: Usable Data Coverage by State and Year



Solid black squares indicate availability of usable speech for given state and year. Empty squares indicate lack of speech data for that year. Some states experience periodicity in missing speeches due to biannual delivery of addresses, such as Texas.

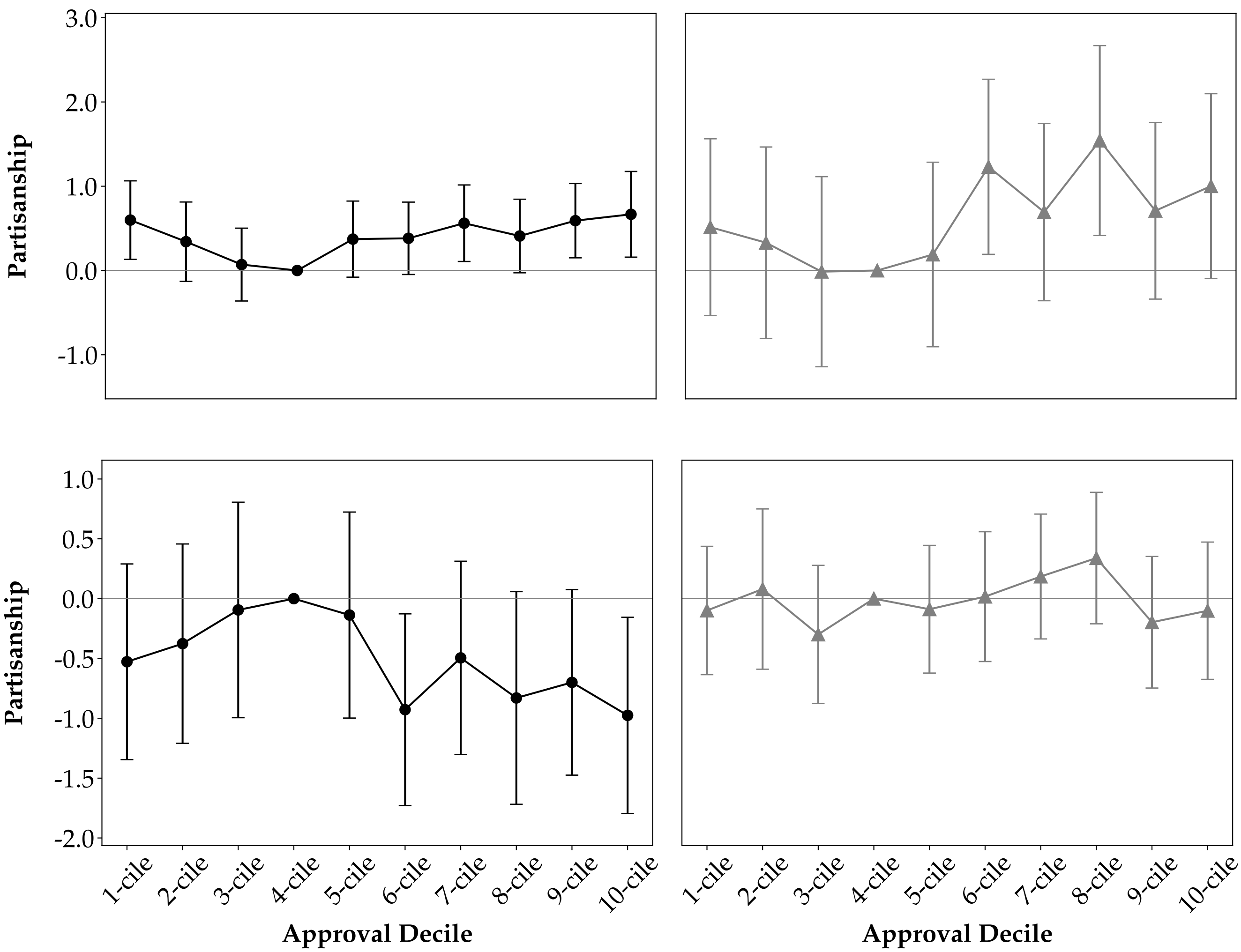
Figure B.2: Partisanship of U.S. Gubernatorial Speeches, Full Speech, Policy Proposals, Past Policies, and Other Speech, vs. Congressional Record: 1990-2020



Partisanship of gubernatorial speech calculated using Gentzkow, Shapiro, and Taddy (2019) leave-out estimator. “Gub. Policy Proposals” computes estimator for gubernatorial speech snippets coded as discussing policy proposals on corpus of U.S. governors’ speech from 1990-2020. “Full Gub. Speech” computes estimator for all gubernatorial speech in given year. “Gub. Past Policies” computes estimator for snippets coded as discussing policies but not policy proposals. “Gub. Other” computes estimator for snippets coded as not discussing policies. “Congress (GST)” series is partisanship of Congressional speech using leave-out estimator, replicating Figure 2A of Gentzkow, Shapiro, and Taddy (2019).

Figure B.3: Unadjusted Levels of Partisanship for Reelectable and Term-Limited Governors by Approval Decile, Aligned vs. Unaligned Legislatures, 1990-2020

(a) Reelectable Govs., Unaligned Legislatures (b) Reelectable Govs., Aligned Legislatures



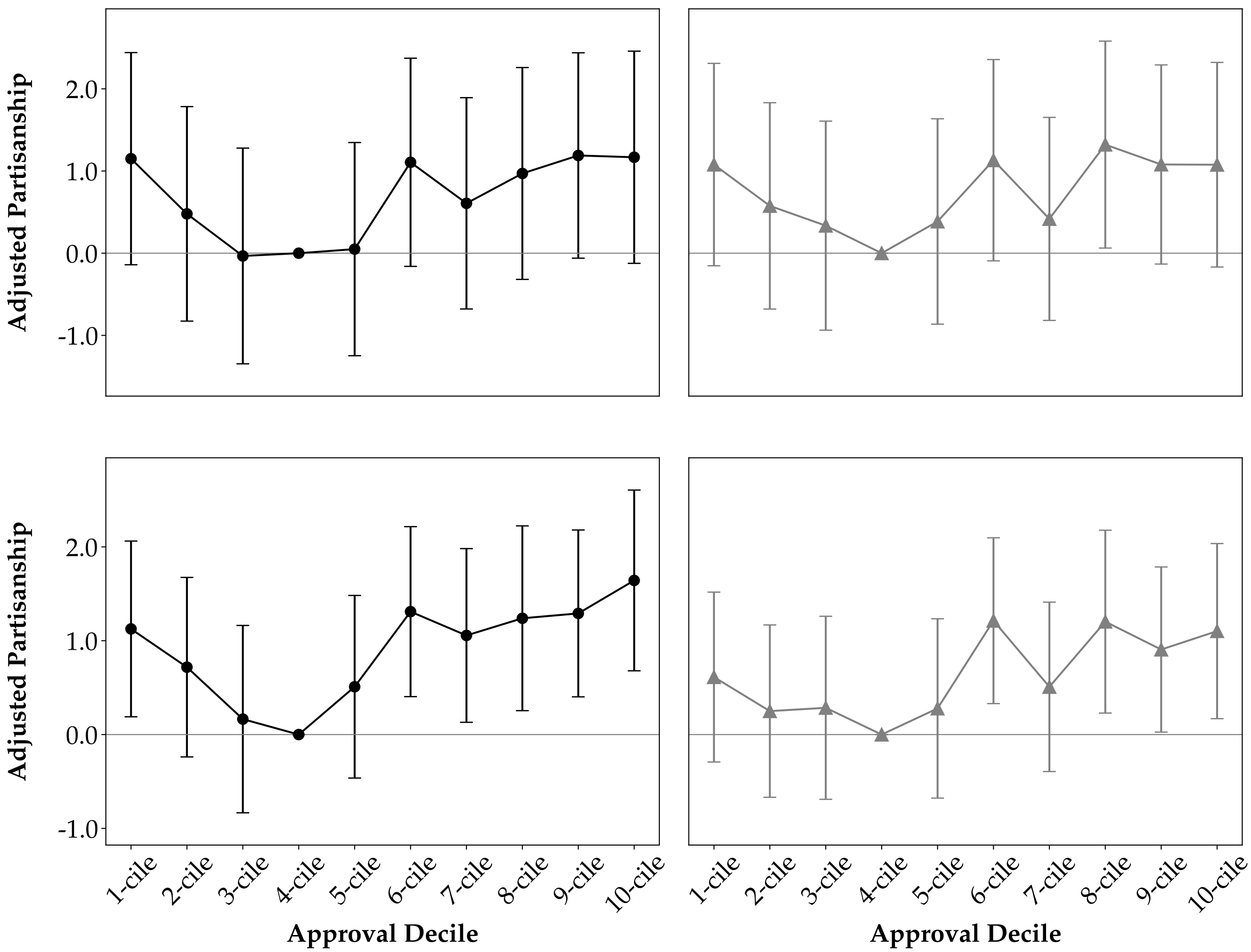
(c) Term-Limited, Unaligned Legislatures (d) Term-Limited, Aligned Legislatures

Dependent variable measures partisanship of U.S. governor State of the State speeches using leave-out estimator of Gentzkow, Shapiro, and Taddy (2019), as calculated in equation (6), normalized by sample mean and standard deviation. Horizontal axis plots approval decile of gubernatorial approval by state, calculated using previous year's fourth quarter approval from Singer (2023) after first year in office, with first quarter approval utilized for first year in office. Panels (a) and (b) plot levels of partisanship for reelectable governors. Panel (a) plots $\alpha_k + \beta_k$ coefficients from equation (9), measuring level of partisanship for reelectable governors with unaligned legislatures in decile k relative to fourth decile. Panel (b) plots $\alpha_k + \beta_k + \gamma_k + \delta_k$ coefficients from equation (9), measuring level of partisanship for reelectable governors with unaligned legislatures in decile k relative to fourth decile. Panels (c) and (d) plot levels of partisanship for term-limited governors. Panel (c) plots α_k coefficients from equation (9), measuring level of partisanship for term-limited governors with unaligned legislatures in decile k relative to fourth decile. Panel (d) plots $\alpha_k + \gamma_k$ coefficients from equation (9), measuring level of partisanship for term-limited governors with unaligned legislatures in decile k relative to fourth decile.

Figure B.4: Partisanship of Reelectable Governors by Approval Decile, Adjusting for Term-Limited Behavior, Varying Definitions of Legislative Alignment, 1990-2020

(a) < 45% Legislators of Gov. Party

(b) ≥ 45% Legislators of Gov. Party

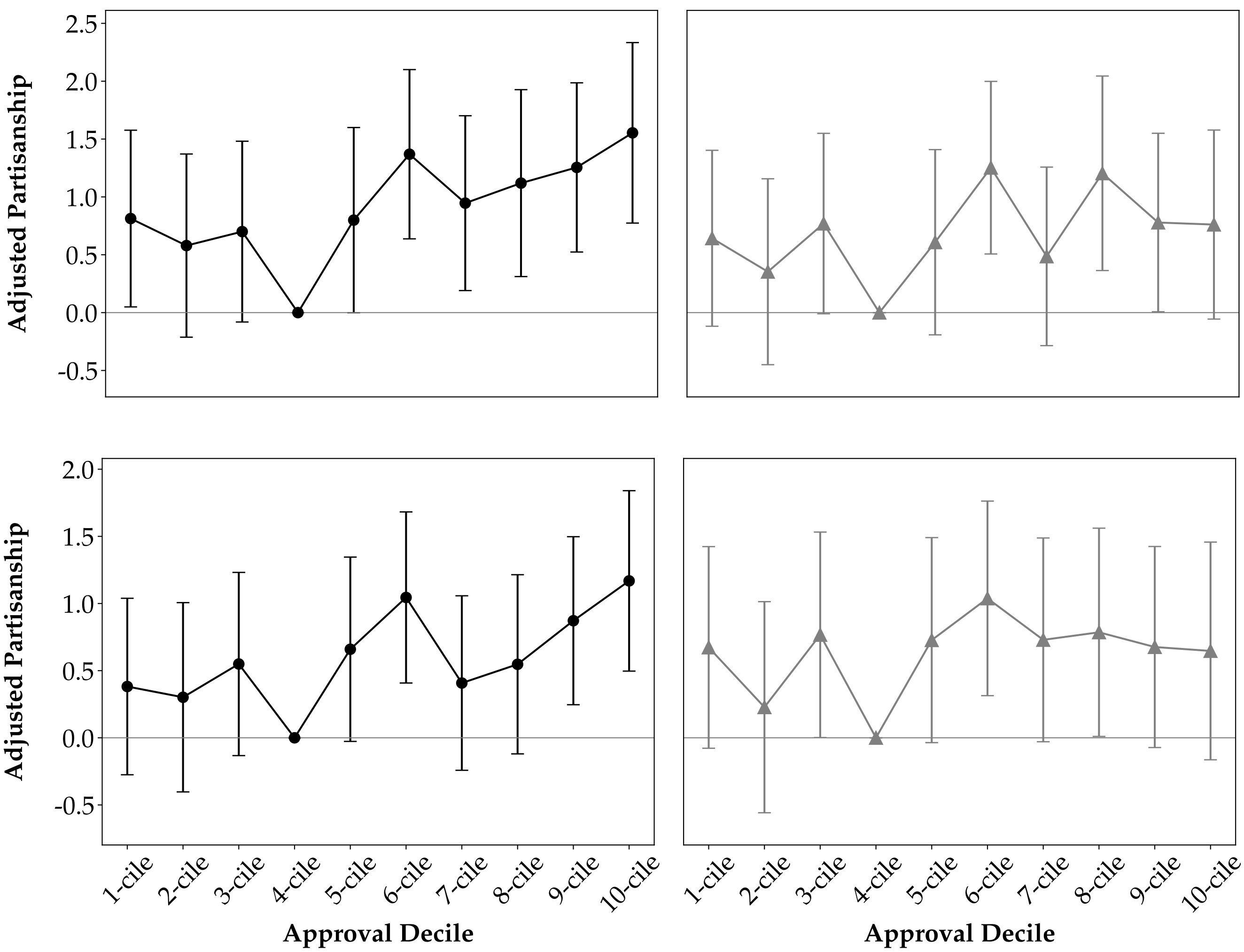


(c) < 50% Legislators of Gov. Party

(d) ≥ 50% Legislators of Gov. Party

(e) < 55% Legislators of Gov. Party

(f) ≥ 55% Legislators of Gov. Party



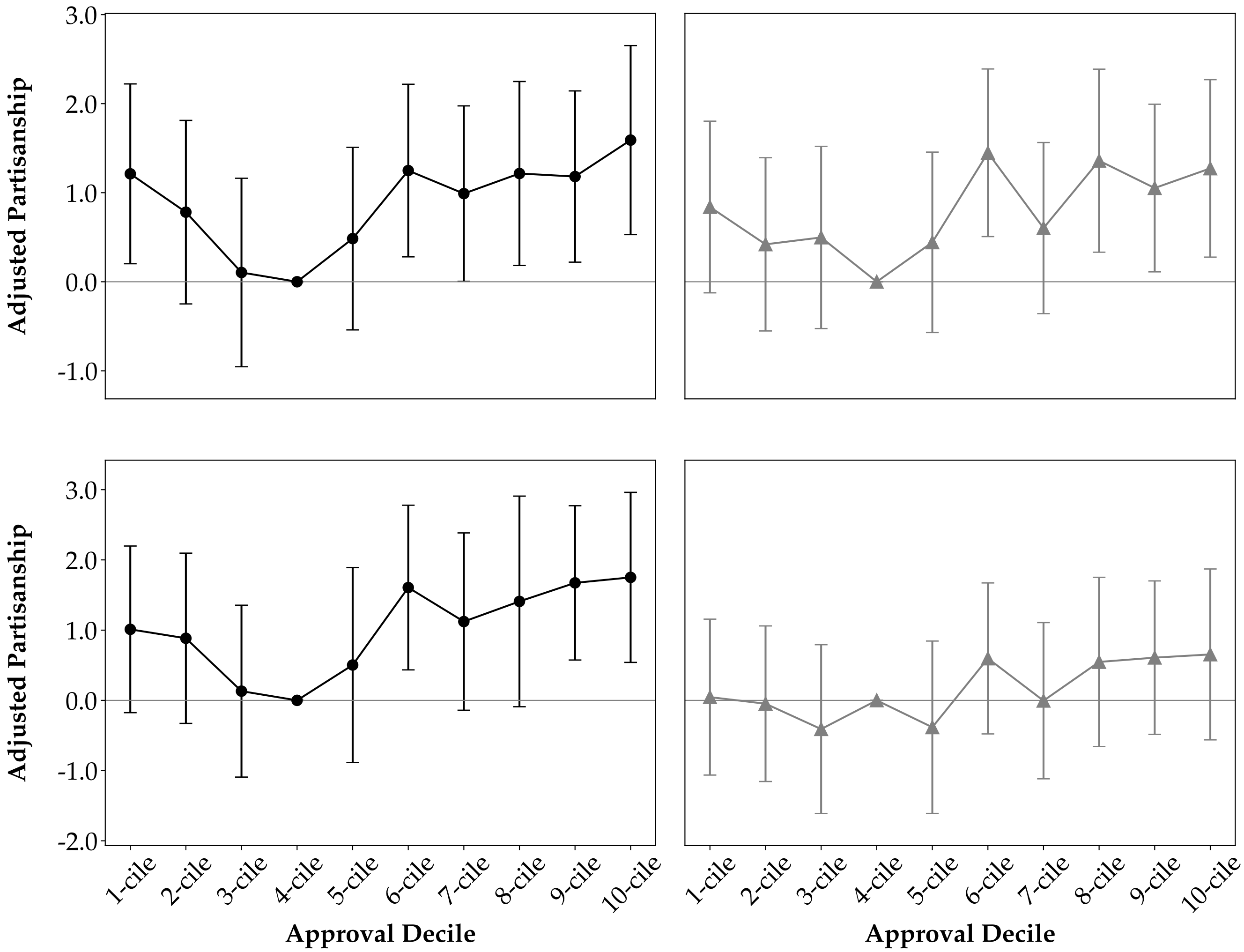
(g) < 60% Legislators of Gov. Party

(h) ≥ 60% Legislators of Gov. Party

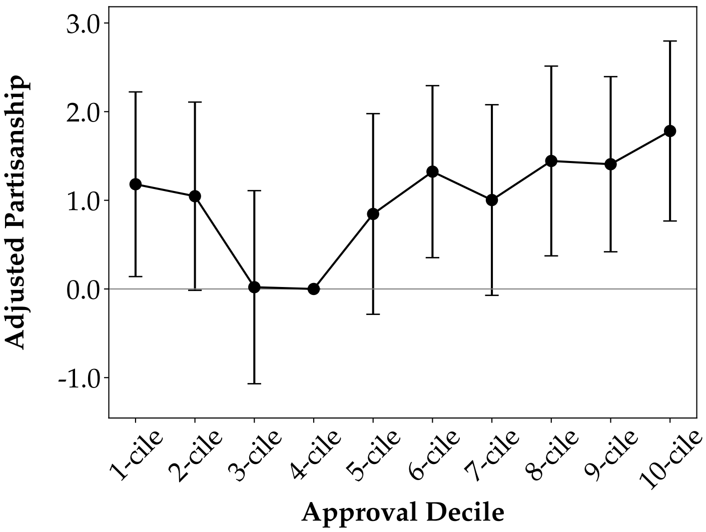
Dependent variable measures partisanship of U.S. governor State of the State speeches using leave-out estimator of Gentzkow, Shapiro, and Taddy (2019), as calculated in equation (6), normalized by sample mean and standard deviation. Horizontal axis plots approval decile of gubernatorial approval by state, calculated using previous year's fourth quarter approval from Singer (2023) after first year in office, with first quarter approval utilized for first year in office. Black circles plots β_k coefficients from equation (9), measuring level of partisanship for reelectable governors with unaligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Gray triangles plot $\beta_k + \delta_k$, measuring level of partisanship for reelectable governors with aligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Bands around coefficient estimates display 95% confidence intervals. Legislative alignment defined as whether more than x% of legislators match governor's party. In panels (a) and (b), x = 45%; in (c) and (d), x = 50%; in (e) and (f), x = 55%, in (g) and (h), x = 60%. Additional controls include vote share for governor's party in most recent presidential election, length of governor's current tenure, state-level unemployment rate, and total state budget expenditures. For panels (a) and (b), N: 1108; num. states: 48; within R^2: 0.07. For panels (c) and (d), N: 1108; num. states: 48; within R^2: 0.06. For panels (e) and (f), N: 1108; num. states: 48; within R^2: 0.07. For panels (e) and (f), N: 1108; num. states: 48; within R^2: 0.07.

Figure B.5: Partisanship of Reelectable Governors by Approval Decile, Adjusting for Term-Limited Behavior, Variation with Electoral Cycle, 1990-2020

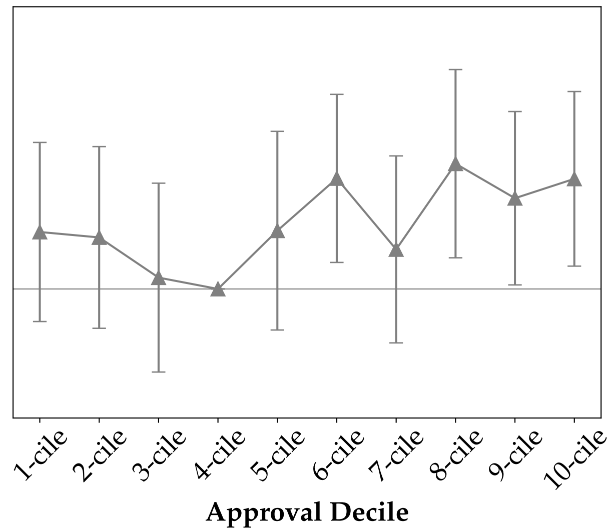
(a) Non-Election Years, Unaligned Leg. (b) Non-Election Years, Aligned Leg.



(c) Election Years Only, Unaligned Leg. (d) Election Years Only, Aligned Leg.



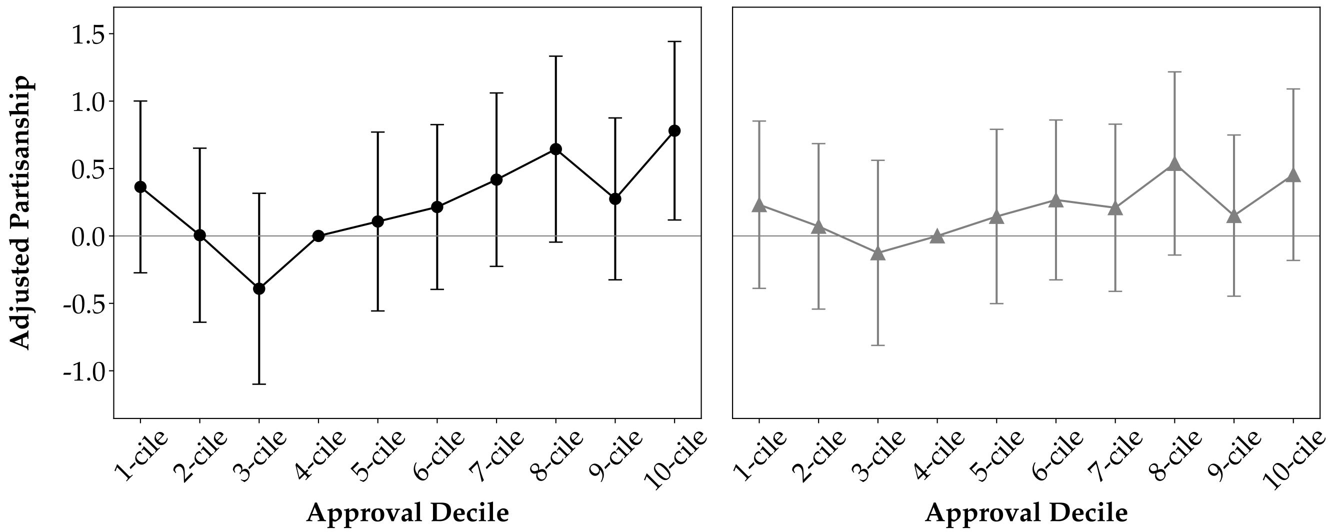
(e) No Term-Limited Last Year, Unaligned Leg.



(f) No Term-Limited Last Year, Aligned Leg.

Dependent variable measures partisanship of U.S. governor State of the State speeches using leave-out estimator of Gentzkow, Shapiro, and Taddy (2019), as calculated in equation (6), normalized by sample mean and standard deviation. Horizontal axis plots approval decile of gubernatorial approval by state, calculated using previous year's fourth quarter approval from Singer (2023) after first year in office, with first quarter approval utilized for first year in office. Black circles plots β_k coefficients from equation (9), measuring level of partisanship for reelectable governors with unaligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Gray triangles plot $\beta_k + \delta_k$, measuring level of partisanship for reelectable governors with aligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Bands around coefficient estimates display 95% confidence intervals. Legislative alignment defined as whether more than half of state legislators match governor's party. Additional controls include vote share for governor's party in most recent presidential election, length of governor's current tenure, state-level unemployment rate, and total state budget expenditures. Panels (a) and (b) drop datapoints where reelectable governors face gubernatorial elections later that year. Panels (c) and (d) drops all datapoints for reelectable governors except governors facing gubernatorial elections later that year. Panels (e) and (f) drop datapoints for term-limited governors in their last full year in office. For panels (a) and (b), N: 910. Num. States: 48. Within R^2: 0.07. For panels (c) and (d), N: 517; num. states: 46; within R^2: 0.12. For panels (e) and (f), N: 1028; num. states: 48; within R^2: 0.07.

Figure B.6: Partisanship of Reelectable Governors by Approval Decile, Adjusting for Term-Limited Behavior, Unaligned vs. Aligned Legislatures, Including Governor Fixed Effects 1990-2020



(a) Unaligned Legislatures

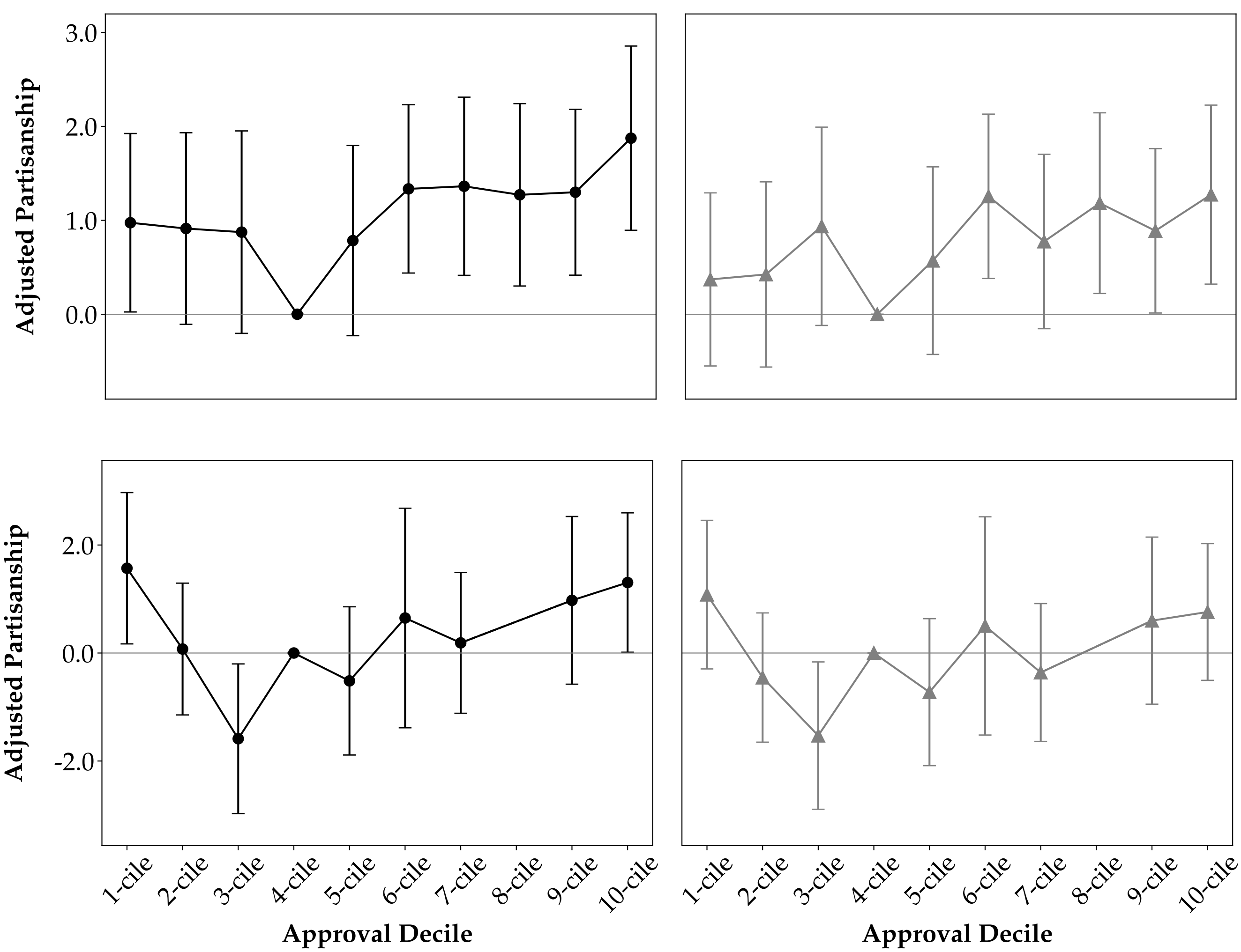
(b) Aligned Legislatures

Dependent variable measures partisanship of U.S. governor State of the State speeches using leave-out estimator of Gentzkow, Shapiro, and Taddy (2019), as calculated in equation (6), normalized by sample mean and standard deviation. Horizontal axis plots approval decile of gubernatorial approval by state, calculated using previous year's fourth quarter approval from Singer (2023) after first year in office, with first quarter approval utilized for first year in office. Estimating equation is identical to (9) but utilizes governor fixed effects instead of state effects. Black circles plots β_k coefficients from equation (9), measuring level of partisanship for reelectable governors with unaligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Gray triangles plot $\beta_k + \delta_k$, measuring level of partisanship for reelectable governors with aligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Bands around coefficient estimates display 95% confidence intervals. Legislative alignment measured as whether more than half of state legislators match governor's party. Additional controls include vote share for governor's party in most recent presidential election, length of governor's current tenure, state-level unemployment rate, and total state budget expenditures. N: 1089; num. governors: 215; within R^2: 0.08. Regression contains year and governor fixed effects.

Figure B.7: Partisanship of Reelectable Governors by Approval Decile, Adjusting for Term-Limited Behavior, Excluding vs. Only Including Term-Limited Governors with Presidential Aspirations, 1990-2020

(a) Dropping Term-Limited Govs. with Presidential Aspirations, Unaligned Legislatures

(b) Dropping Term-Limited Govs. with Presidential Aspirations, Aligned Legislatures



(c) Dropping Term-Limited Govs. without Presidential Aspirations, Unaligned Legislatures

(d) Dropping Term-Limited Govs. without Presidential Aspirations, Aligned Legislatures

Dependent variable measures partisanship of U.S. governor State of the State speeches using leave-out estimator of Gentzkow, Shapiro, and Taddy (2019), as calculated in equation (6), normalized by sample mean and standard deviation. Horizontal axis plots approval decile of gubernatorial approval by state, calculated using previous year's fourth quarter approval from Singer (2023) after first year in office, with first quarter approval utilized for first year in office. Black circles plots β_k coefficients from equation (9), measuring level of partisanship for reelectable governors with unaligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Gray triangles plot $\beta_k + \delta_k$, measuring level of partisanship for reelectable governors with aligned legislatures in decile k, adjusting for behavior of term-limited governors, relative to fourth decile. Term-limited governors in panels (a) and (b) drop any governors who eventually ran for U.S. president or were nominated for vice president. Term-limited governors in panels (c) and (d) include only governors who eventually ran for U.S. president or were nominated for vice president. Data missing for 8–ciles in panels (c) and (d). Bands around coefficient estimates display 95% confidence intervals. Legislative alignment measured as whether more than half of state legislators match governor's party. Additional controls include vote share for governor's party in most recent presidential election, length of governor's current tenure, state-level unemployment rate, and total state budget expenditures. For panels (a) and (b), N: 843. Num. States: 48. Within R^2: 0.08. For panels (c) and (d), N: 1053; num. states: 48; within R^2: 0.08.

Table B.1: Standard Deviations of Partisanship, Governors vs. Congress, 1990-2020

Standard Deviation	1990-1999	2000-2009	2010-2020
Overall	0.053	0.031	0.041
Within-State	0.041	0.027	0.029
Within Governor	0.028	0.016	0.017
N	375	482	479
Num. Governors	80	92	104
Overall	0.015	0.019	0.028
Within-State	0.014	0.018	0.027
Within-Congressman	0.008	0.012	0.017
N	2,701	2,686	3,261
Num. Congressmen	906	779	1,030

Partisanship of gubernatorial speech calculated using Gentzkow, Shapiro, and Taddy (2019) leave-out estimator on U.S. gubernatorial speech snippets coded as discussing policy proposals. Partisanship of Congressional speech replicated from Gentzkow, Shapiro, and Taddy (2019) using leave-out estimator. 1990-1999 utilizes Congresses 101-105, 2000-2009 Congresses 106-110, 2010-2020 Congresses 111-116.

Table B.2: Partisanship by Approval Decile, Legislative Alignment, and Reelection Eligibility, 1990-2020

		Reelect ×	Leg. Align ×	Reelect × Leg. Align ×
Appr. 1-cile	-0.527	1.126**	0.428	-0.513
	(0.417)	(0.477)	(0.497)	(0.311)
Appr. 2-cile	-0.376	0.718	0.456	-0.468
	(0.424)	(0.487)	(0.545)	(0.310)
Appr. 3-cile	-0.094	0.165	-0.205	0.120
	(0.459)	(0.509)	(0.542)	(0.294)
Appr. 5-cile	-0.137	0.509	0.049	-0.231
	(0.439)	(0.496)	(0.516)	(0.310)
Appr. 6-cile	-0.928**	1.310***	0.945*	-0.096
	(0.408)	(0.462)	(0.496)	(0.291)
Appr. 7-cile	-0.495	1.056**	0.680	-0.548*
	(0.412)	(0.472)	(0.496)	(0.310)
Appr. 8-cile	-0.830*	1.239**	1.169**	-0.036
	(0.453)	(0.502)	(0.538)	(0.305)
Appr. 9-cile	-0.700*	1.291***	0.503	-0.385
	(0.395)	(0.453)	(0.490)	(0.308)
Appr. 10-cile	-0.975**	1.642***	0.874	-0.540
	(0.418)	(0.491)	(0.513)	(0.345)
Constant	0.302	-0.832**	-0.259	0.320*
	(0.452)	(0.380)	(0.391)	(0.220)
N			1108	
Num. States			48	
Year F.E.			Yes	
State F.E.			Yes	
R^2 Within			0.07	

Dependent variable measures partisanship of U.S. governor State of the State speeches using leave-out estimator of Gentzkow, Shapiro, and Taddy (2019), as calculated in equation (6), normalized by sample mean and standard deviation. Approval decile represents decile of approval rating by state, calculated using previous year's fourth quarter approval from Singer (2023) after first year in office, with first quarter approval utilized for first year in office. First column of coefficients measures baseline levels of partisanship by approval decile for term-limited governors, i.e. α_k terms in equation (9), with constant corresponding to omitted category of fourth decile. Names of second through fourth columns represent variables being interacted with approval decile. Second column of coefficients measures baseline partisanship of reelectable relative to term-limited governors by approval decile, i.e. β_k coefficients. Third column measures level of partisanship for legislatively aligned term-limited governors relative to unaligned, i.e. γ_k coefficients. Fourth column measures level of partisanship for reelectable governors with aligned legislatures relative to unaligned legislatures, i.e. δ_k coefficients. Third column compares partisanship of reelectable governors in gubernatorial election years to all term-limited governors. Fourth column compares partisanship of reelectable governors to all term-limited governors except those in their last year in office. Additional controls (estimates omitted) include vote share for governor's party in most recent presidential election, length of governor's current tenure, state-level unemployment rate, and total state budget expenditures. Data for Idaho (no approval data) and Nebraska (does not recognize political parties in legislature) are dropped. Standard errors in parentheses. $^{*}p < 0.1, ^{**}p < 0.05, ^{***}p < 0.01$.

Table B.3: Swing States, Republican States, and Democratic States

Swing States	Republican States	Democratic States
AK	AL	CO
AR	AZ	DE
CA	FL	HI
CT	IA	KY
GA	ID	MD
IL	MA	MO
IN	MI	NC
KS	MS	NY
LA	ND	OR
ME	NE	PA
MN	NM	VA
MT	NV	VT
NH	OH	WA
NJ	SC	WV
OK	SD	
RI	TX	
TN	UT	
WY	WI	

Appendix C Hand-Coding Guide

Below is the guide provided to research assistants for the hand-coding task.

General remark. Remember that the goal of this task is to identify (past or current) policy proposals among gubernatorial speech snippets — as well as risky policy proposals —that will be used to train a large-language-model. As a guiding principle when coding, it may be useful to ask yourself: is the language in this snippet relevant to identifying policy proposals, past/current proposals, or risky policy proposals?

For example, if a governor spends a lot of time in a snippet on rhetoric, but then at the very end mentions a policy she passed, we wouldn't code that as "yes, this is about as policy" even though a policy may be mentioned by name at the end. This is because the language of that snippet, by and large, does not talk about past policy proposals.

If a governor is clearly reflecting on the content of a policy proposal —but the policy is not mentioned by name —this would also be coded as "referring to a policy proposal." The reason is that we are trying to figure out how much time governors spend in their speeches discussing policy proposals (as opposed to other things). So the relevancy of snippets to this category — or any of the other categories —should be assessed using these sorts of heuristics.

Coding Guidelines. The outline below details each of the main categories to be coded, as well as examples ("easy" and "hard") of each of the codings.

1. "Policy." Coded as "1" if the snippet discusses the enactment of a state-level policy (either passed by the governor, state government, or referendum) and "0" if it does not. A policy discussion is a reference to a specific act of legislation or law, a concrete proposal to increase or decrease funding to a certain cause, other legal orders proposed by the government to take certain concrete actions, and discussions of details of any of the above.

    - Example of 0 (easy): "I will continue to speak out against those who promote prejudice. I know you will too. And I will tell you this: a handful of people who may want to burn a cross are no match for 10,000 Idahoans who marched to support the Table Rock Cross." (ID, 2000).
    - Example of 0 (hard): "It is simply not pono for our families to be living in cars, people to be sleeping in the doorways of businesses downtown or on picnic tables in our parks. There is no one silver bullet to solve the problems of homelessness and affordable housing, but there are many good ideas that can and should be enacted." (HI, 2006). Discusses an issue and hints at the concept of a solution, but does not concretely address a policy.

- Example of 0 (hard): "The budget is balanced but great risks and uncertainties lie ahead. The federal government, the courts or changes in the economy all could cost us billions and drive a hole in the budget. The ultimate costs of expanding our health care system under the Affordable Care Act are unknown. Ignoring such known unknowns would be folly, just as it would be to not pay down our wall of debt. That is how we plunged into a decade of deficits." (CA, 2013). Does not actually discuss a governor or state-led policy initiative, despite referring to the ACA (a federal initiative).
- Example of 1 (easy): "The Reform Albany Act will have as its centerpiece an independent ethics commission that will have jurisdiction over State government. This commission will have the power to enforce campaign finance and end pay-to-play and bring jurisdiction and oversight to so-called good government groups, who hide their donors behind walls of sanctimony." (NY, 2010)
- Example of 1 (hard): "We were asked to meet yet another list of requirements. The federal government objects not on a scientific basis, but upon a vaguely defined legal risk analysis. This is not just about semantics. It is about achieving wolf delisting on rational terms that work for Wyoming. I do not care what we call them as long as we can manage them. The new demands from the federal government go far beyond the word predator and include changing how a pack is defined and even questioning whether the national parks will assume responsibility for half of the 15 packs it plans for Wyoming." (WY, 2004). Refers to federal policy/definitions, which is confusing, but also details of how the state will implement such a policy out in the context of their own state.

2. "Proposal/Past." Only applies if "policy" coded as "1." Coded as "1" if the snippet refers to a policy that has just put into place or will be put into place in the future. Coded as "0" otherwise — in particular, if a governor is reflecting on the effects of a policy in the past. Coded as "0.5" if it contains substantive elements of both. Continuing a preexisting policy implemented from the past without any substantive changes also does not constitute a (future) policy proposal —this would be coded as a "0."

   - Example of 0 (easy): "Clearly we are doing things right. We are making progress. But the job numbers are only part of the story. In addition to making it easier for businesses to create jobs, we have also invested in public works projects. In doing so, we improved our public infrastructure, made it more attractive for businesses to relocate or stay here, and directly created even more jobs." (OR, 2006). Clearly refers to a past policy action but not a future action.
   - Example of 0 (hard): "... We have budgeted more than 260 million for higher ed capital. That funds new science facilities at Jackson State Community College and the University of Tennessee. It also includes nearly 25 million for improvements to our colleges of applied technology all across the state, and it includes the funds to complete the long

awaited fine arts building at East Tennessee State University. The reason we continue to make these investments in education is we want Tennesseans to have the education, training and skills necessary to have a good paying, high-quality job…" (TN, 2015). Suggests that we "have budgeted" (not will budget) money, and that this process will continue, but does not propose something inherently novel.

- Example of 1 (easy): "By reducing our dependence on foreign energy sources we can not only stop sending our energy dollars to unstable parts of the world, but we can become a world leader in clean energy technologies, from wind and solar power to geothermal and fuel cells…." (NY, 2004). References clear future policy action (reducing dependence on foreign energy source.)
- Example of 1 (hard): "We need a new, more dynamic, economic development strategy. One that can leverage the resources of our business sector, as well as higher education and not for profits. The Delaware Economic Development Office needs to be at the forefront of moving Delaware into the 21st century economy. So my first act as governor was to find a way to energize our economic development efforts. We are going to do that by bringing private sector involvement into DEDO." (DE, 2017). Reflects on a past action, but lays out a policy for the coming year (private sector involvement).
- Example of 0.5: "Some said that Louisiana could not change its stripes and make a new start…but we did.And now…after our recent successes in ethics reform and tax reform…we must take the next step forward…an overhaul of our workforce development system." (LA, 2008). Second sentence reflects on past risky policies, and then talks about workforce development.